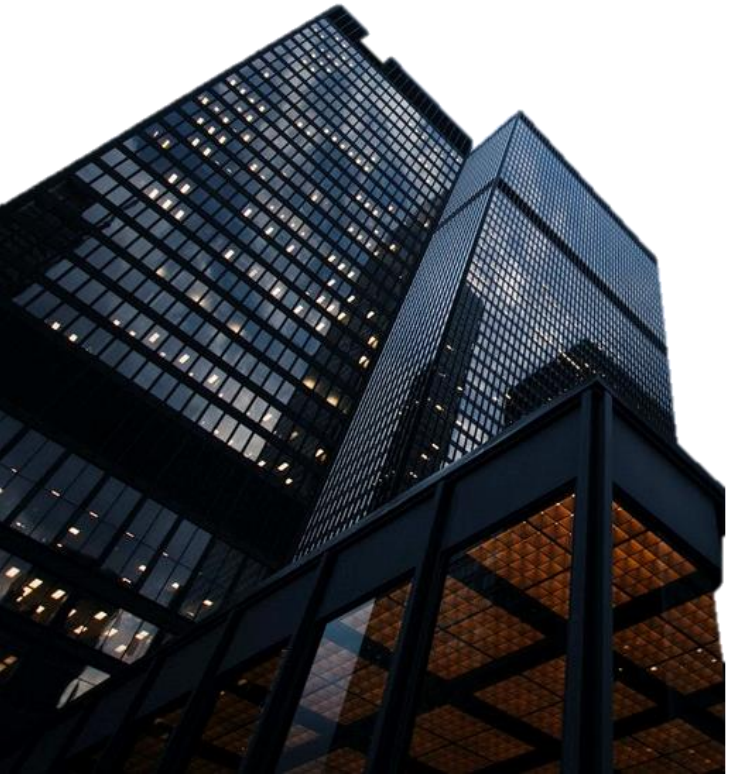
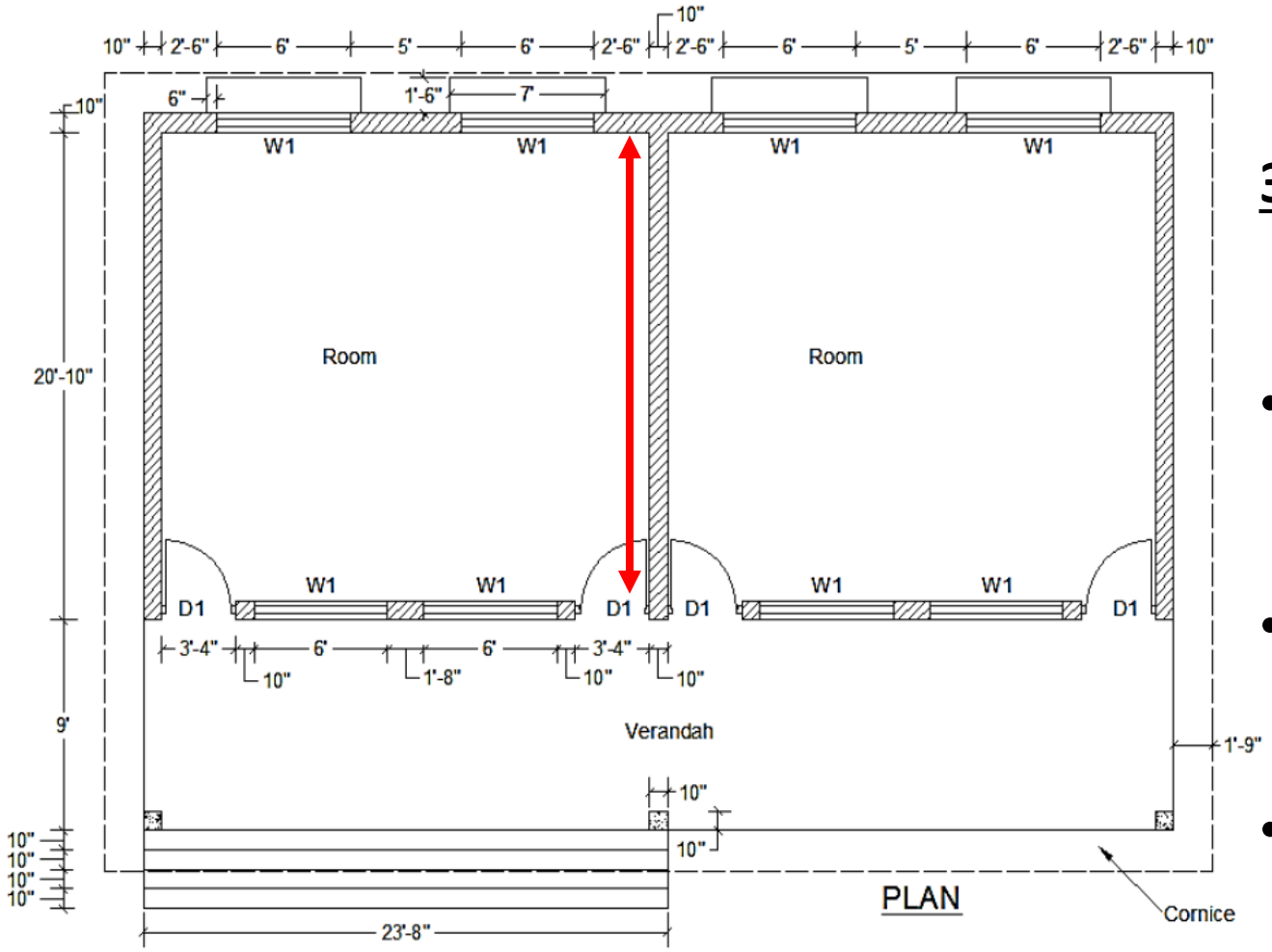
CE 208
Quantity Surveying

Estimation of a Residential Masonry Building



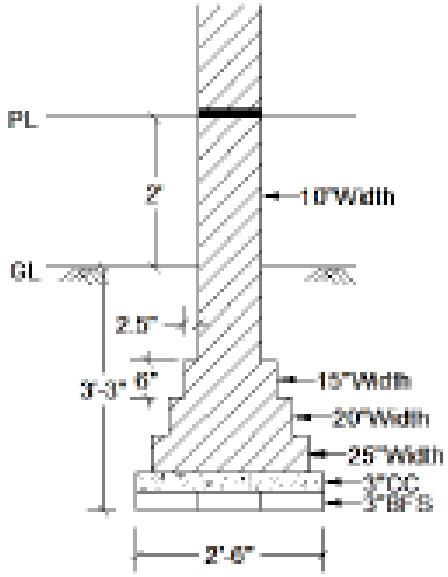
DEPARTMENT OF CIVIL ENGINEERING
BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

5. Estimation of Brickwork from GL to PL

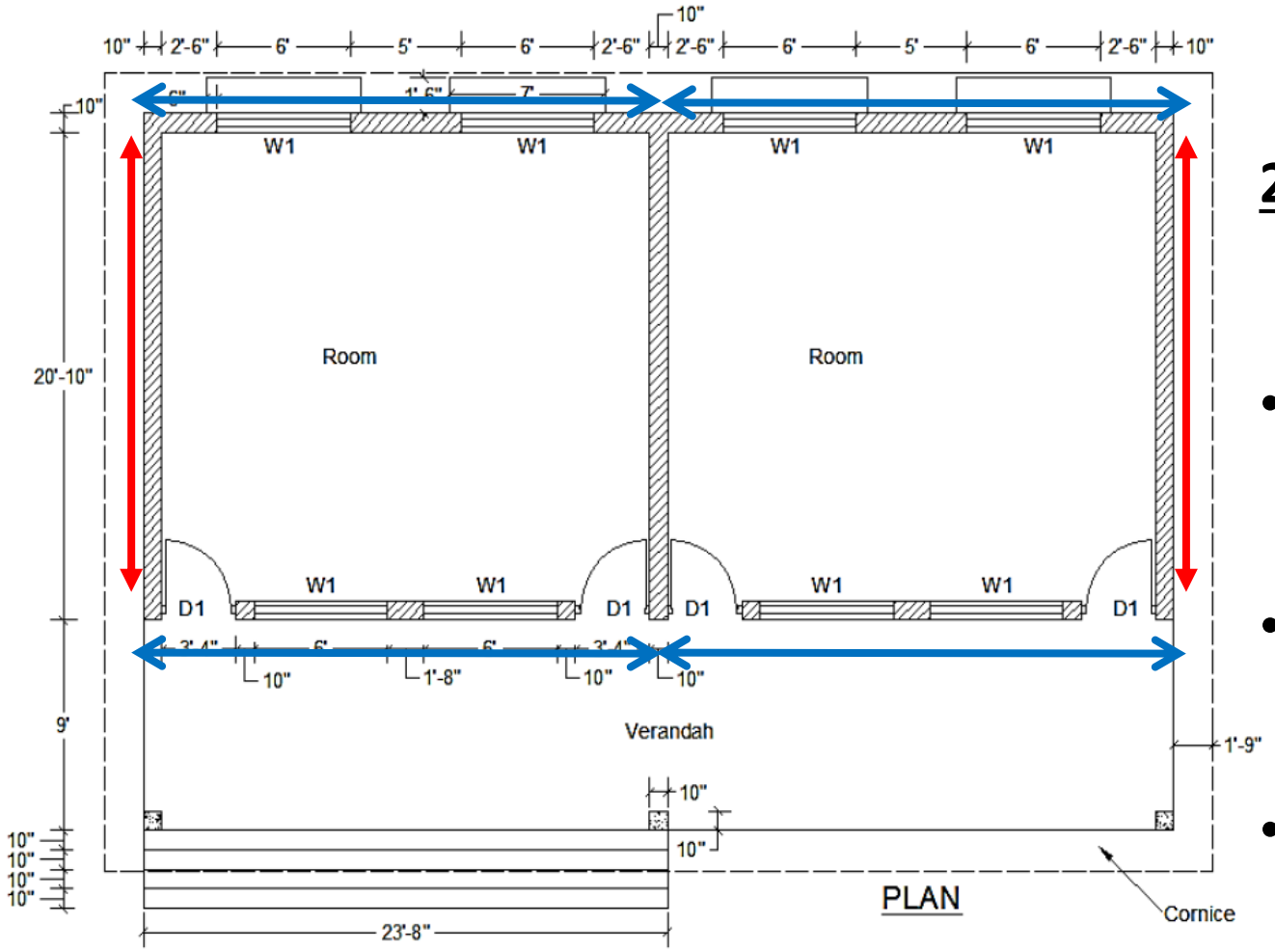


30” wall foundation:

- Height= 2’
- Width= 10”
- Length= 20’10”-10”

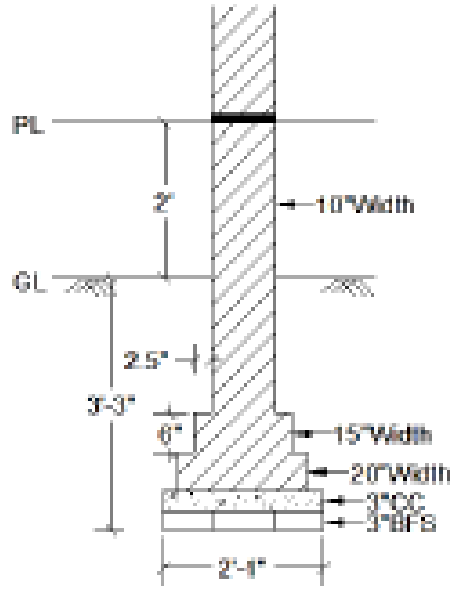


5. Estimation of Brickwork from GL to PL

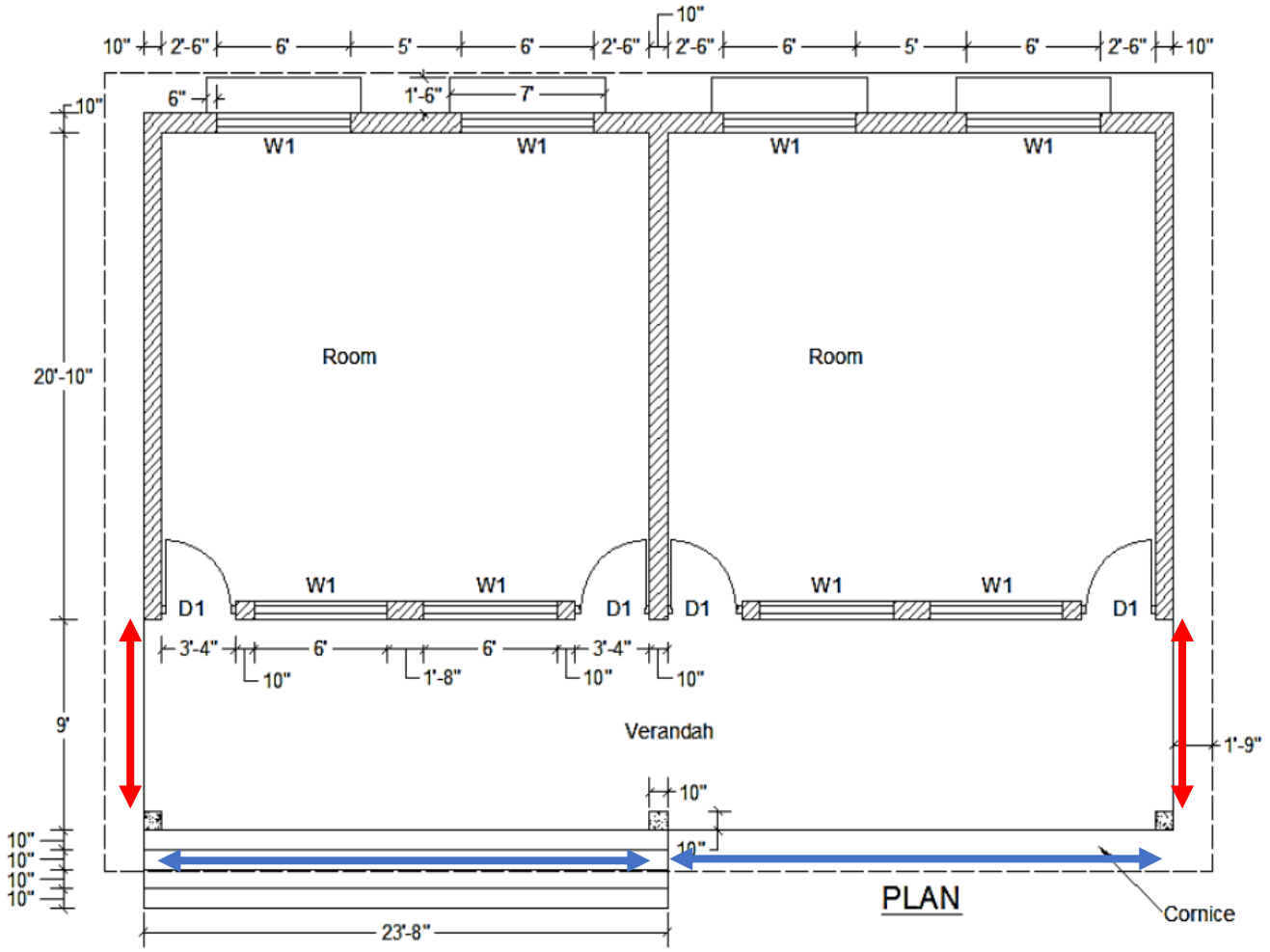


25” wall foundation:

- Height= 2’
- Width= 10”
- Length= (20’10”-10”)*2 +(22’10”+5”)*4



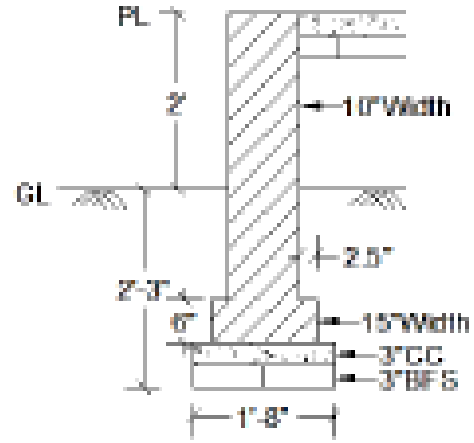
5. Estimation of Brickwork from GL to PL



20” wall foundation:

Height= 2’

Width= 10”



Length= (9’-10”)*2+(22’10”-10”)*2

6. RCC in Column Footing (up to GL)

- Base Slab

Volume= (4’x4’x1’)*3

- Column

Volume= (1’x1’x1’9”)*3

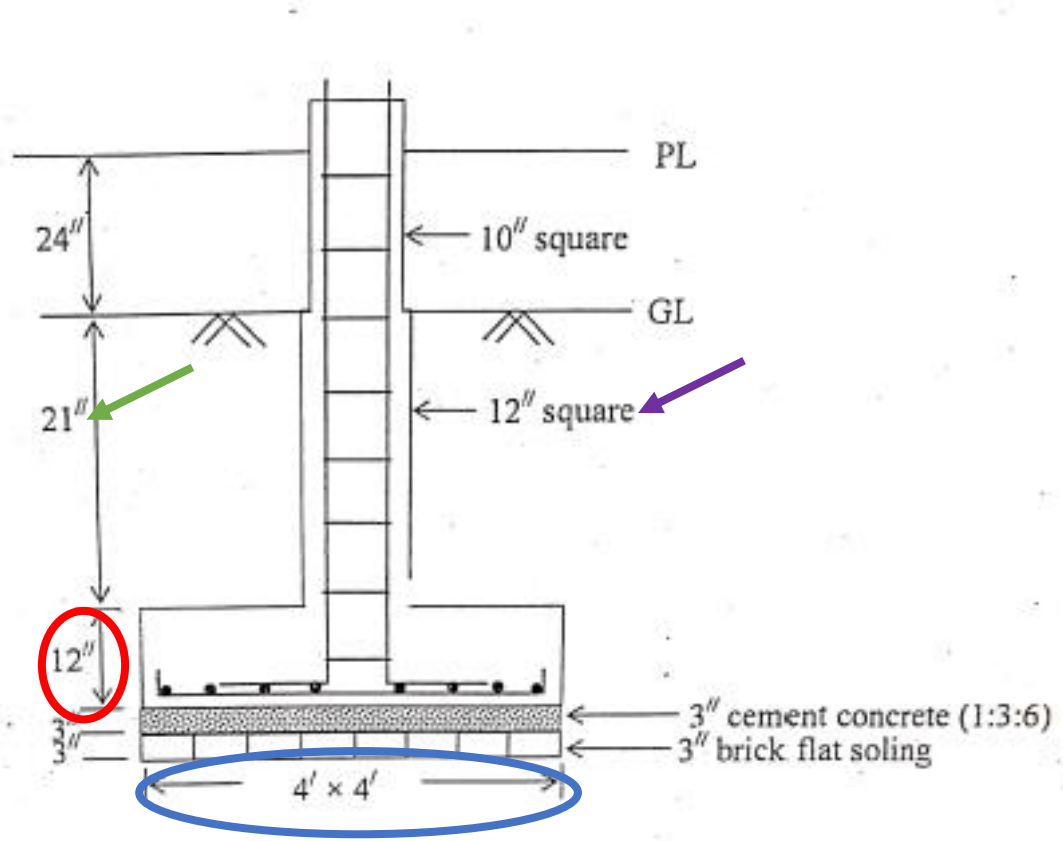


Figure: 4 ft x 4 ft isolated column footing

7. RCC in Column (GL to PL)

- Height= 2’
- Area= 10”x10”
- Volume= (2’x10”x10”)*3

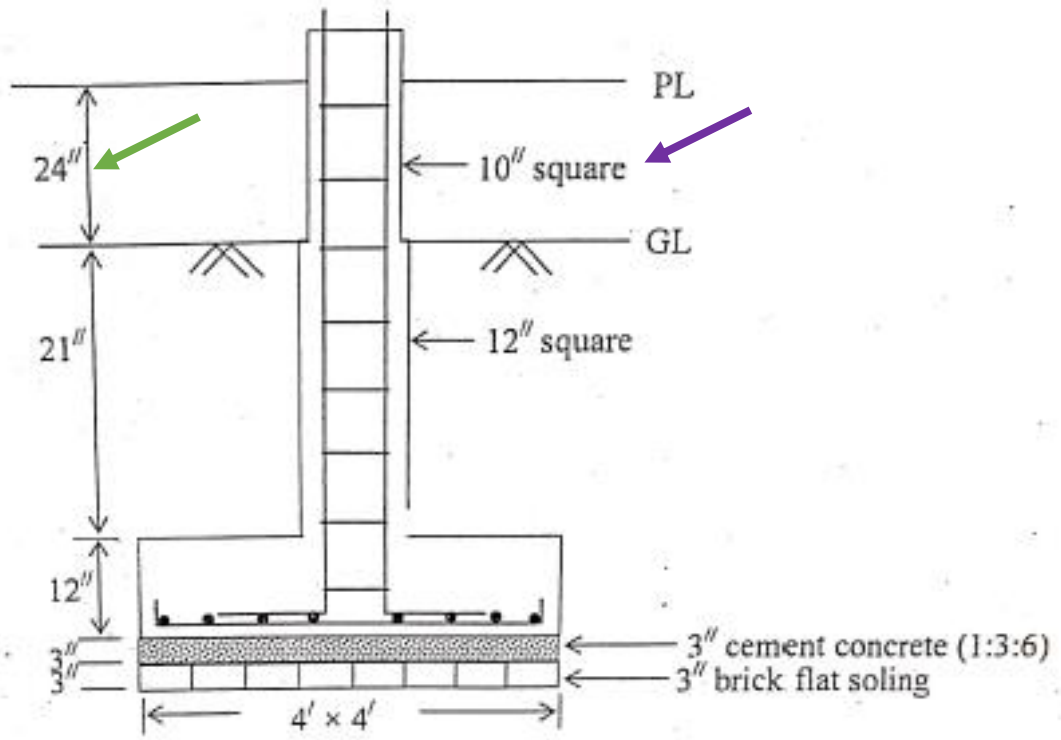


Figure: 4 ft x 4 ft isolated column footing

8. Damp Proof Course (DPC)

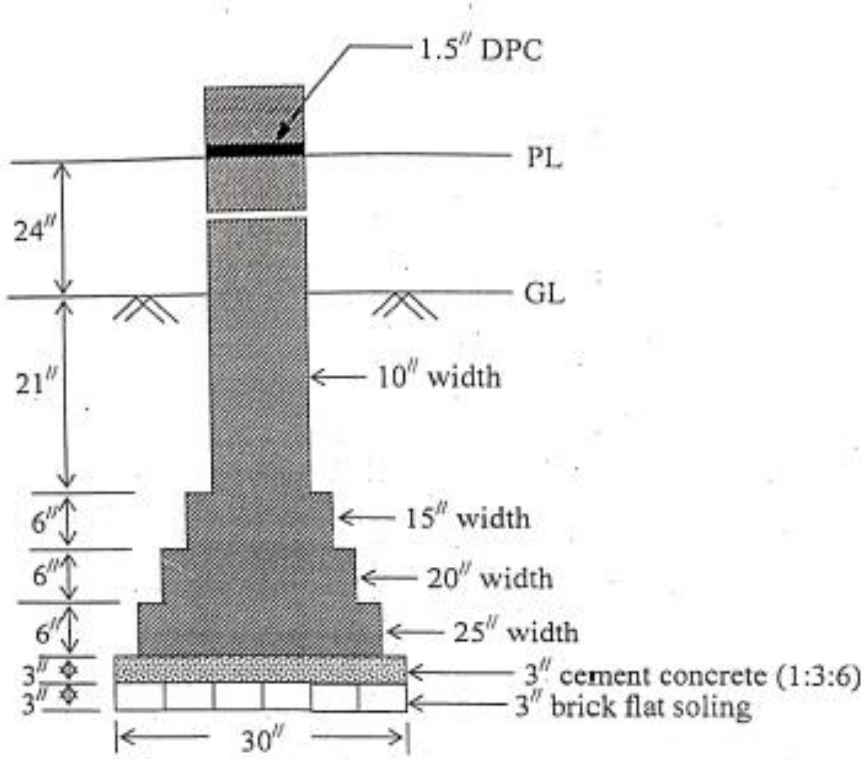


Figure: 30 inch continuous wall foundation

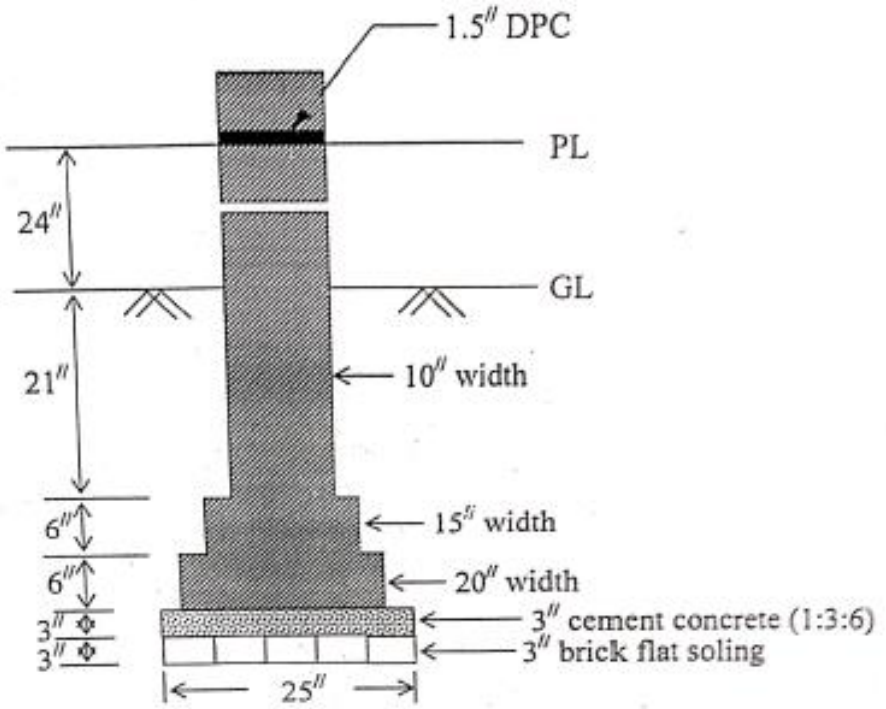


Figure: 25 inch continuous wall foundation

8. Damp Proof Course (DPC)

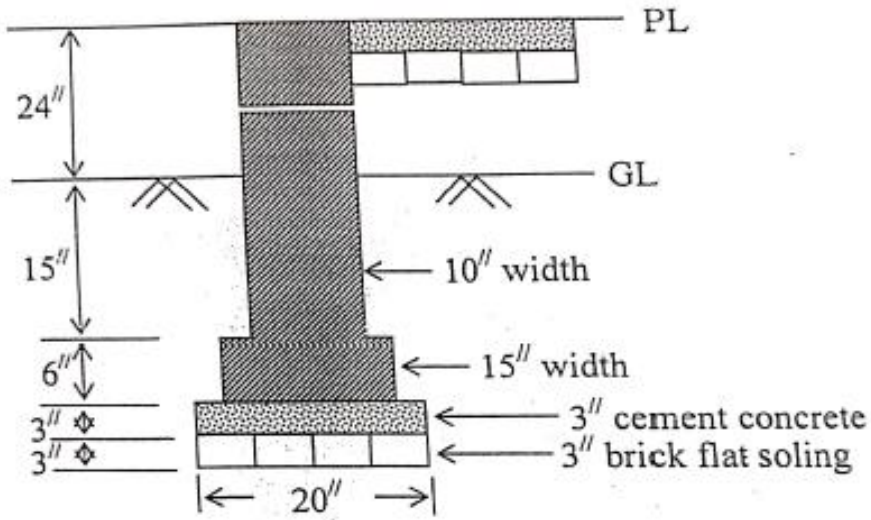


Figure: 20 inch continuous verandah wall foundation

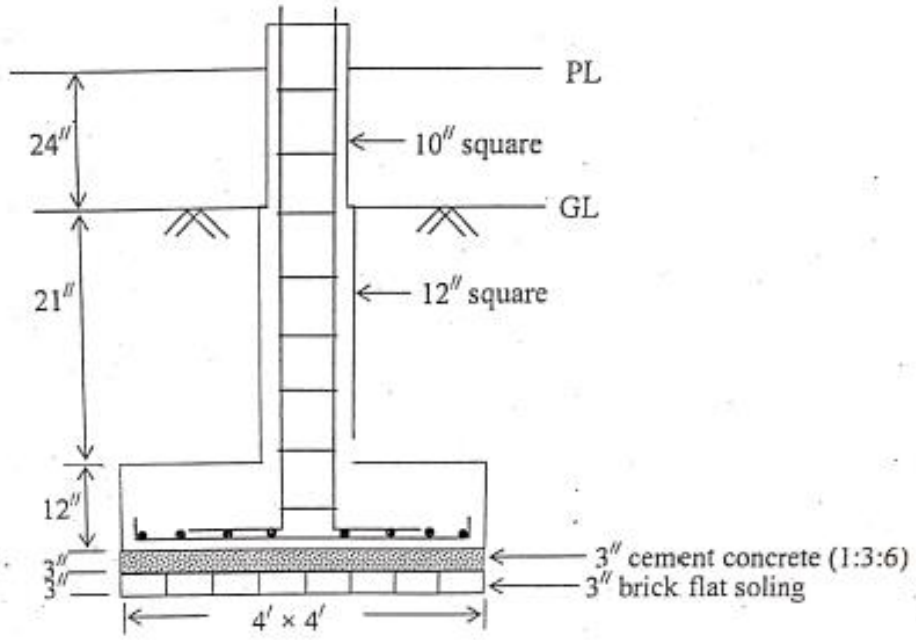
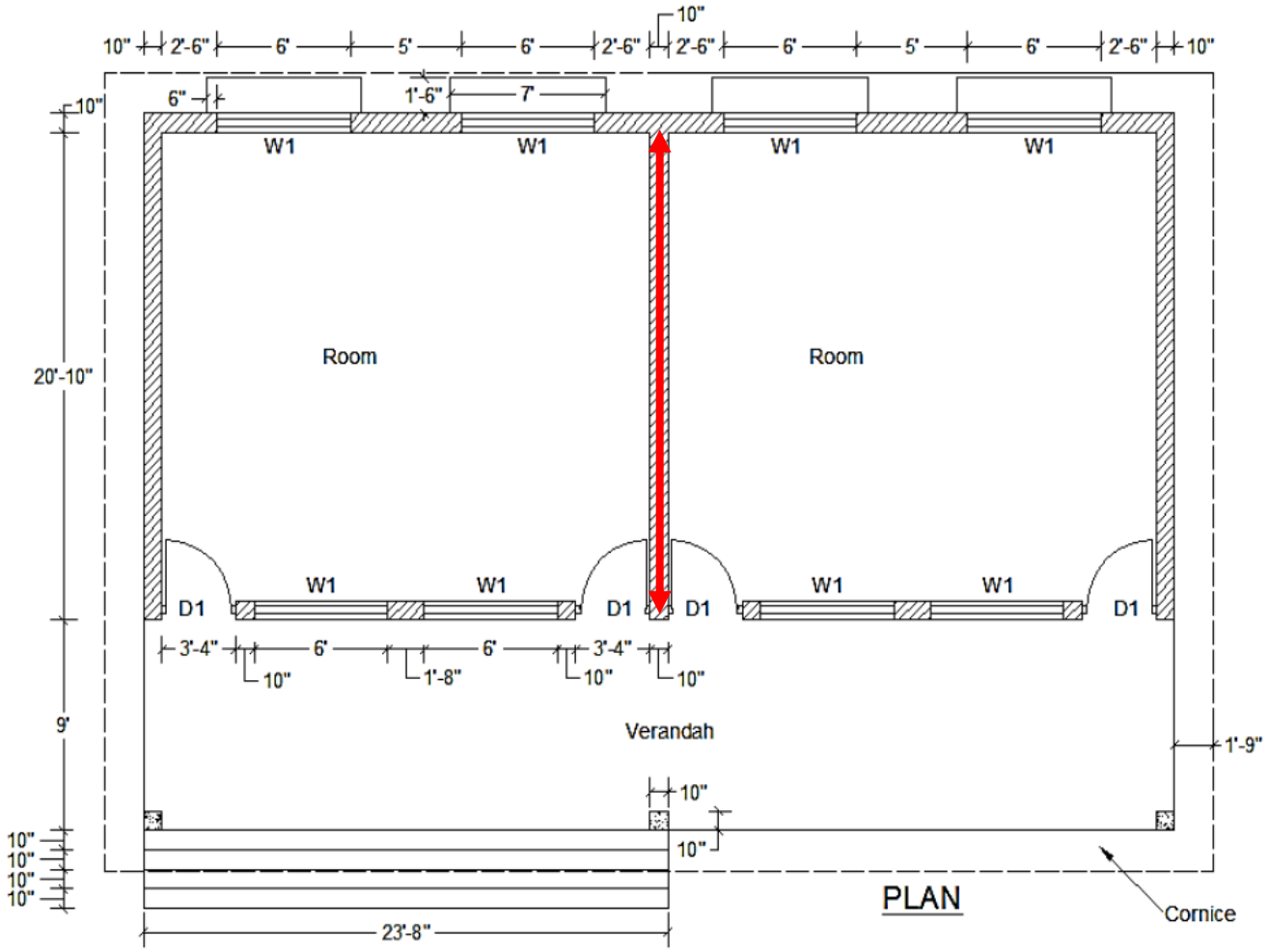


Figure: 4 ft x 4 ft isolated column footing

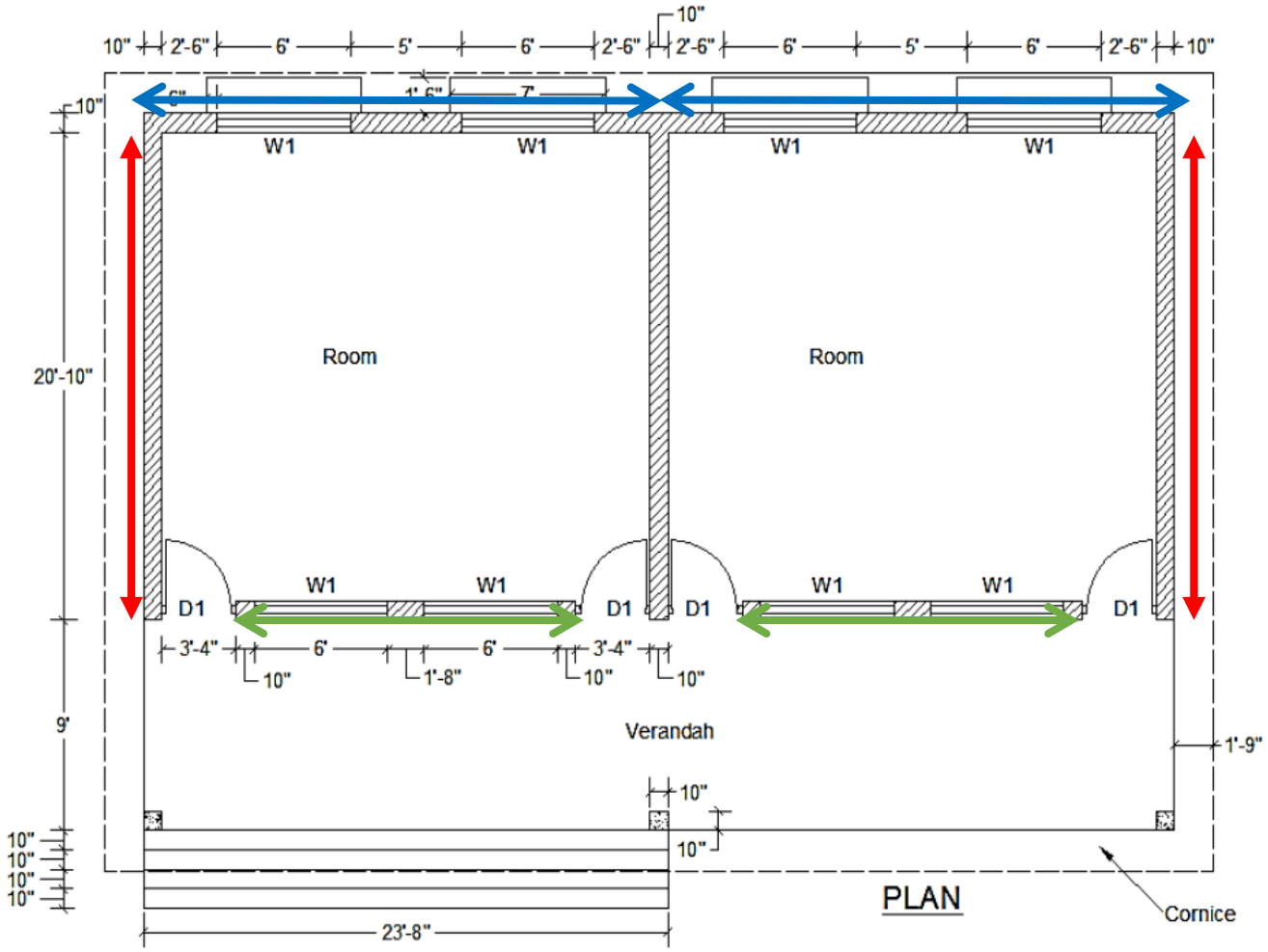
8. Damp Proof Course (DPC)



30” wall foundation:

- Thickness= 1.5”
- Width= 10”
- Length= 20’10”

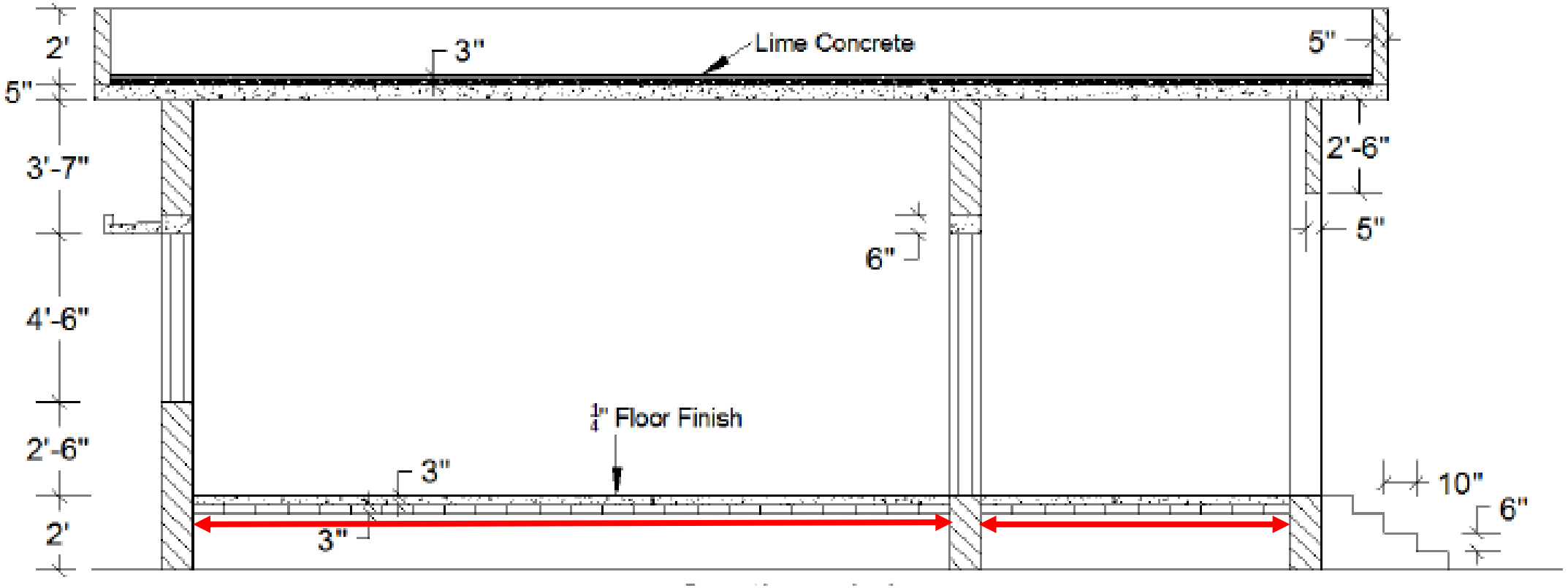
8. Damp Proof Course (DPC)



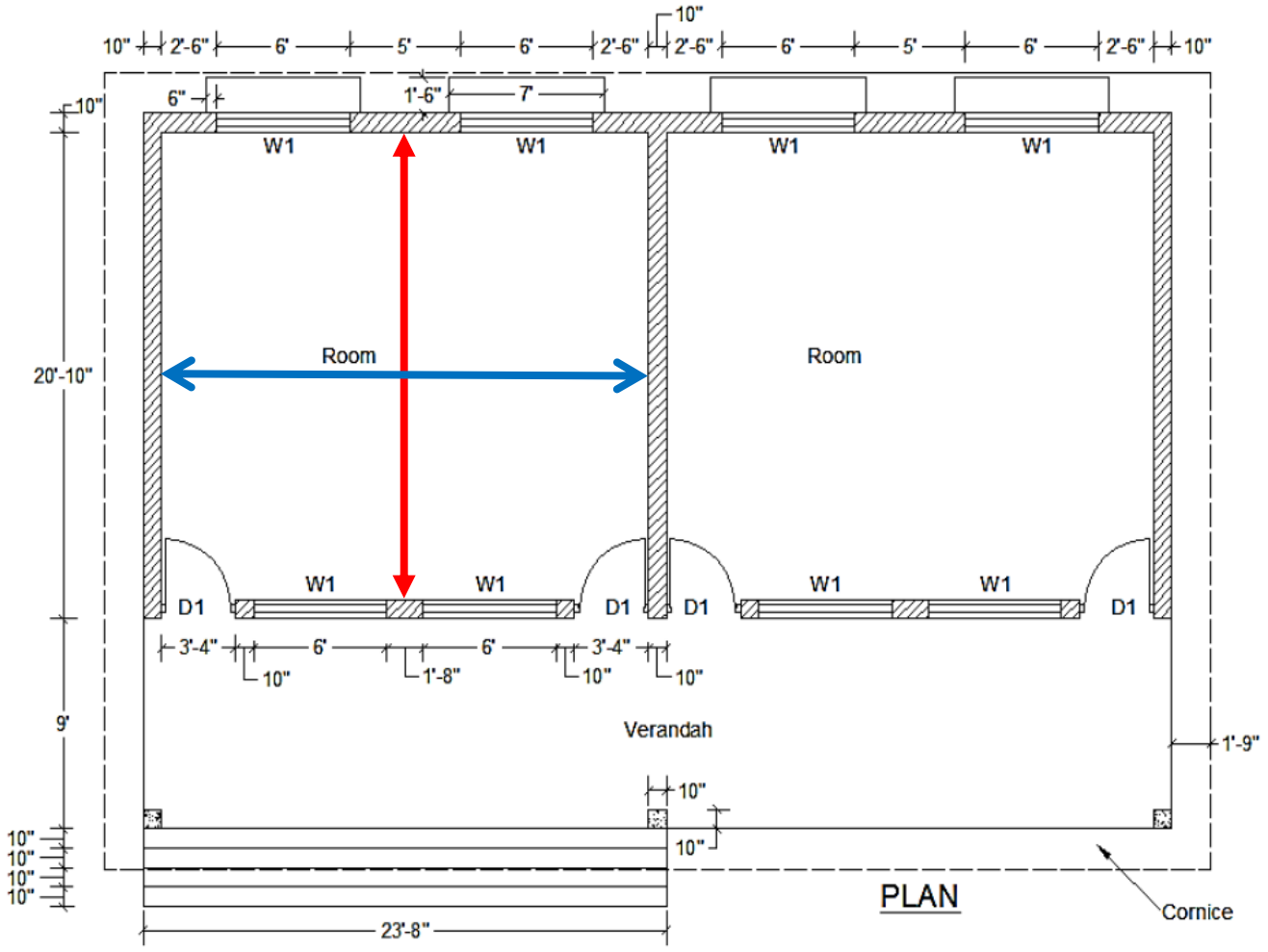
25" wall foundation:

- Thickness= 1.5"
- Width= 10"
- Length= 20'10"*2+(22'10"+5")*2+(15'4")*2

9. BFS (one layer) in Floors



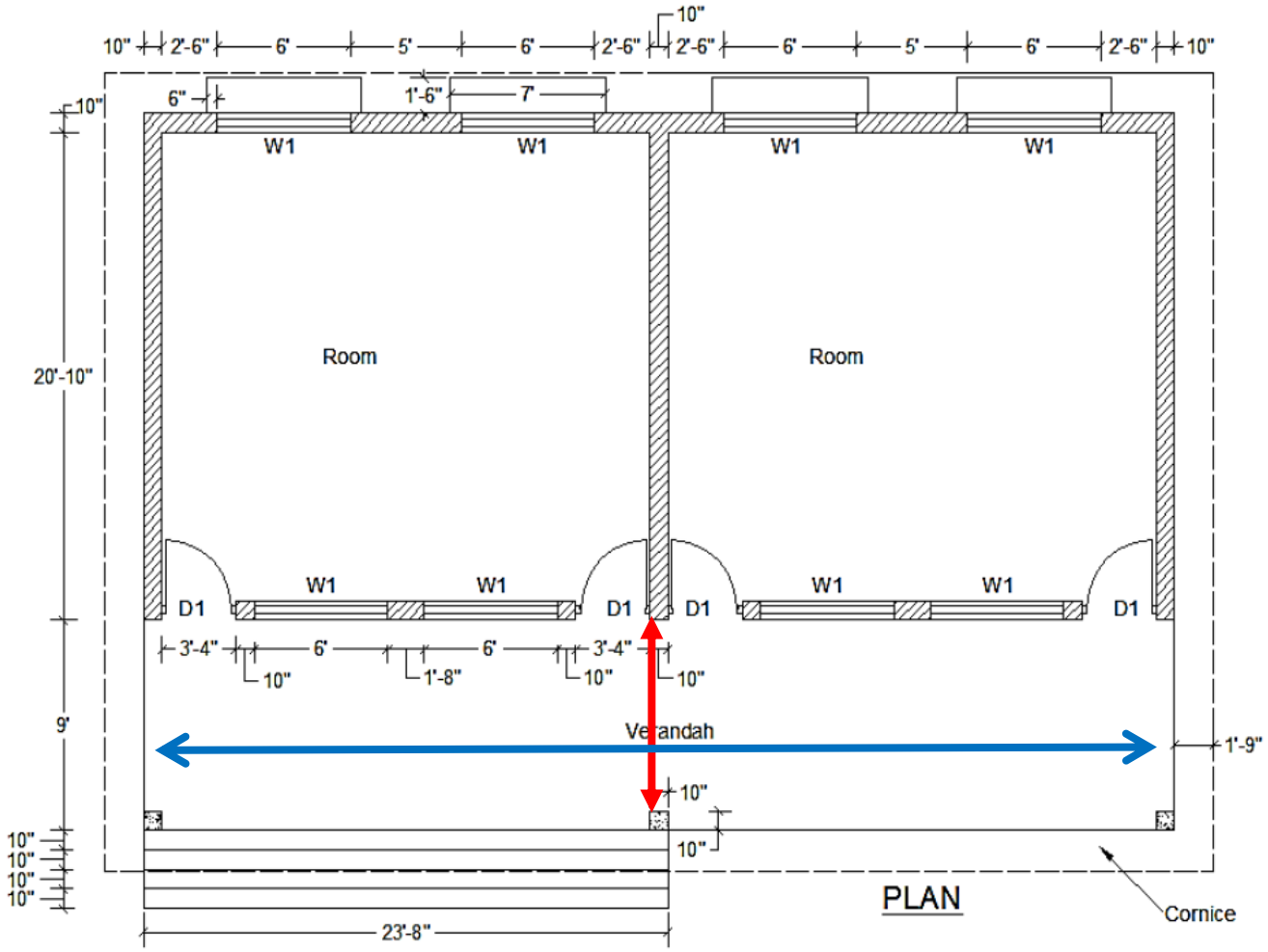
9. BFS (one layer) in Floors



Main Room:

- Length= 22’
- Width= 20’

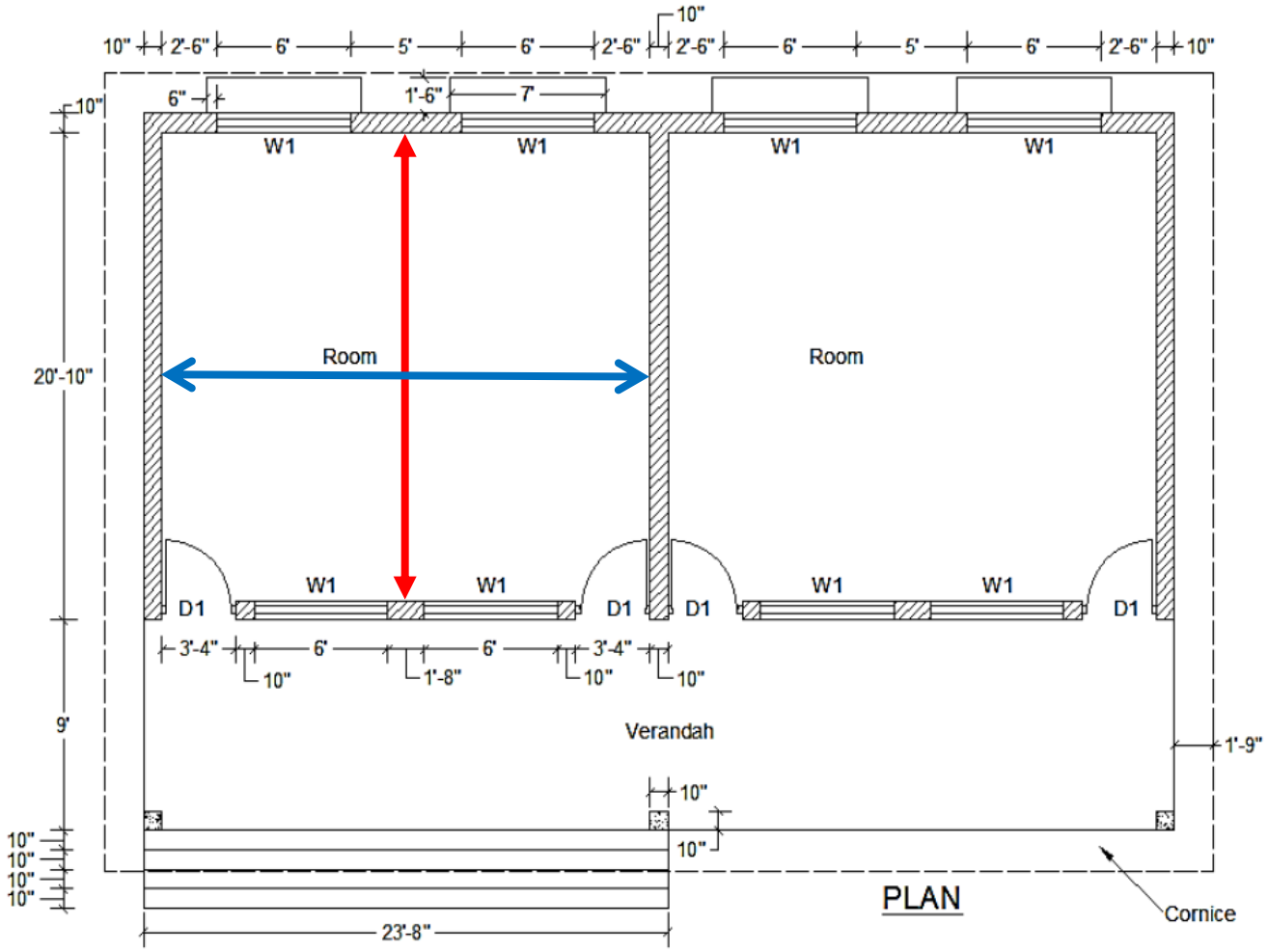
9. BFS (one layer) in Floors



Verandah:

- Length= (22’10”)*2-10”
- Width= 9’-10”

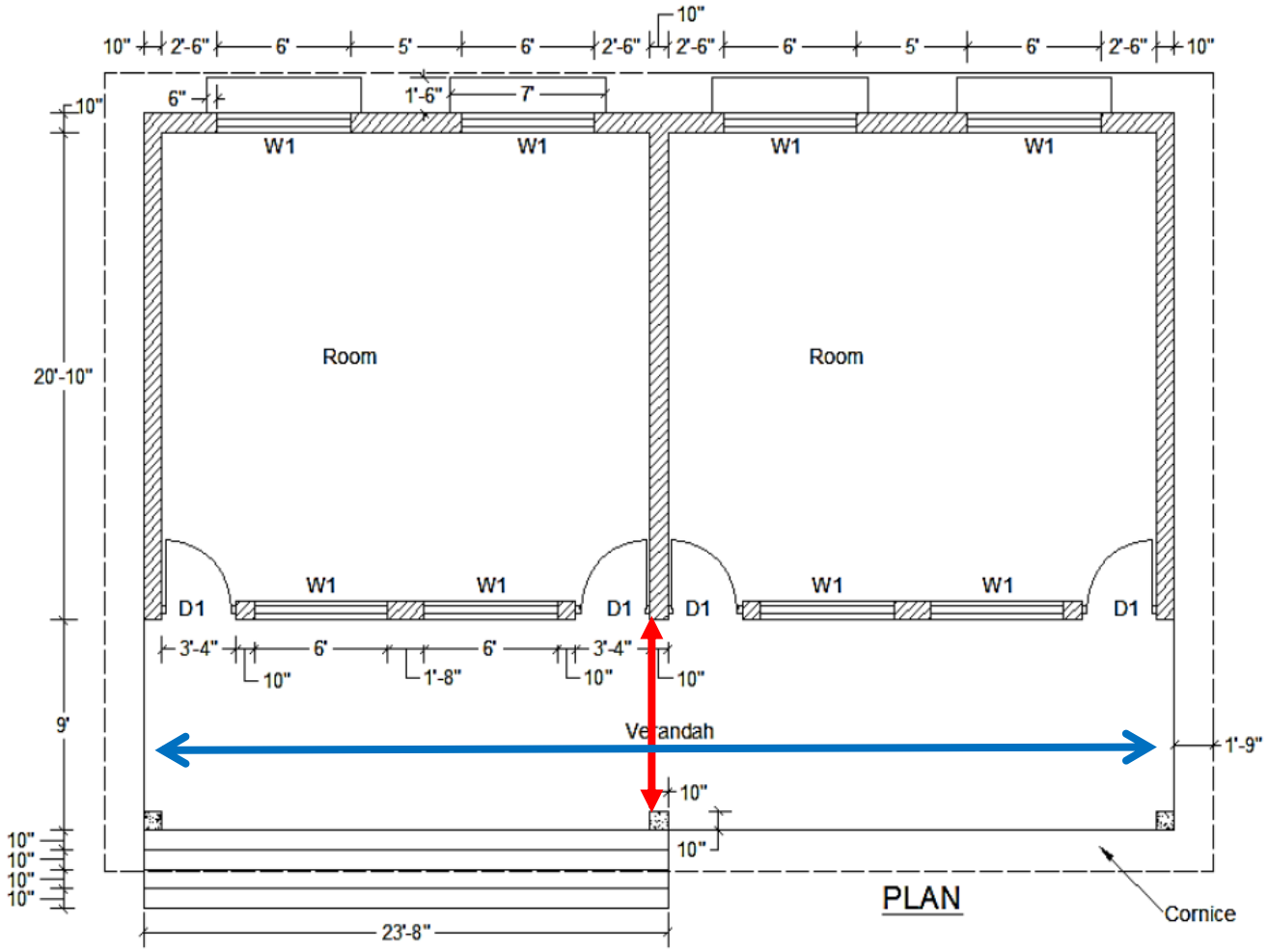
10. CC (3”) in Floors



Main Room:

- Length= 22’
- Width= 20’
- Thickness= 3”

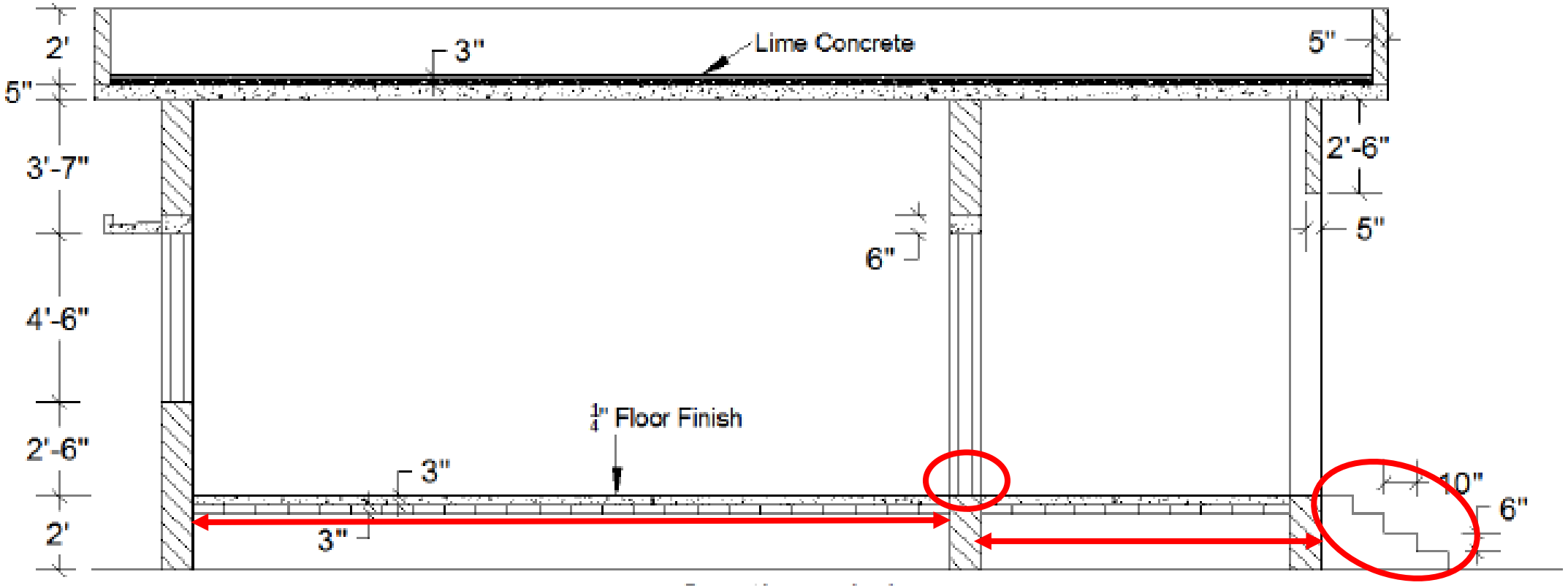
10. CC (3”) in Floors



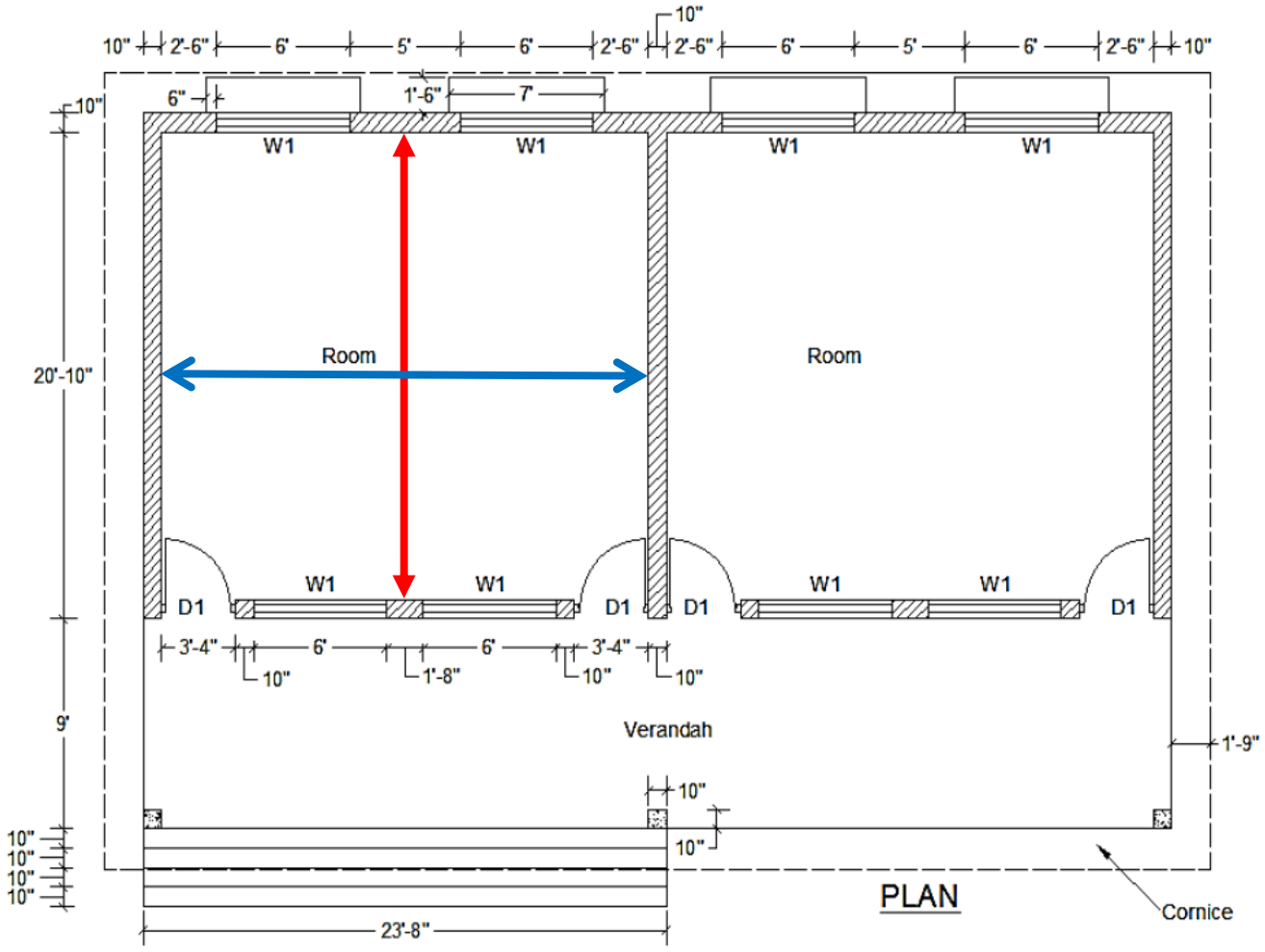
Verandah:

- Length= (22’10”)*2-10”
- Width= 9’-10”
- Thickness= 3”

11. Floor Finish, ¼” thick



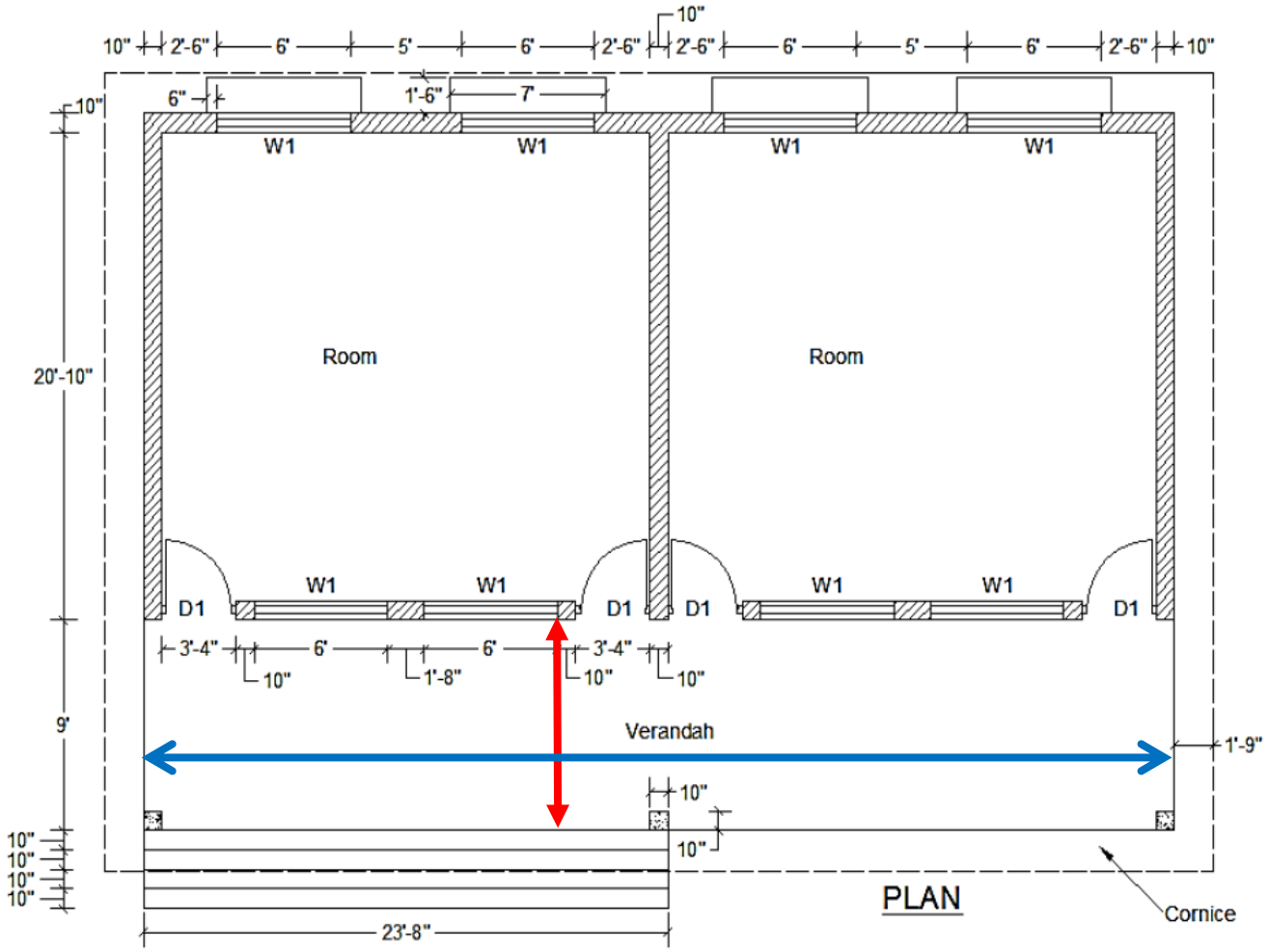
11. Floor Finish, ¼” thick



Main Floor:

- Length= 22’
- Width= 20’
- Thickness= 1/4”

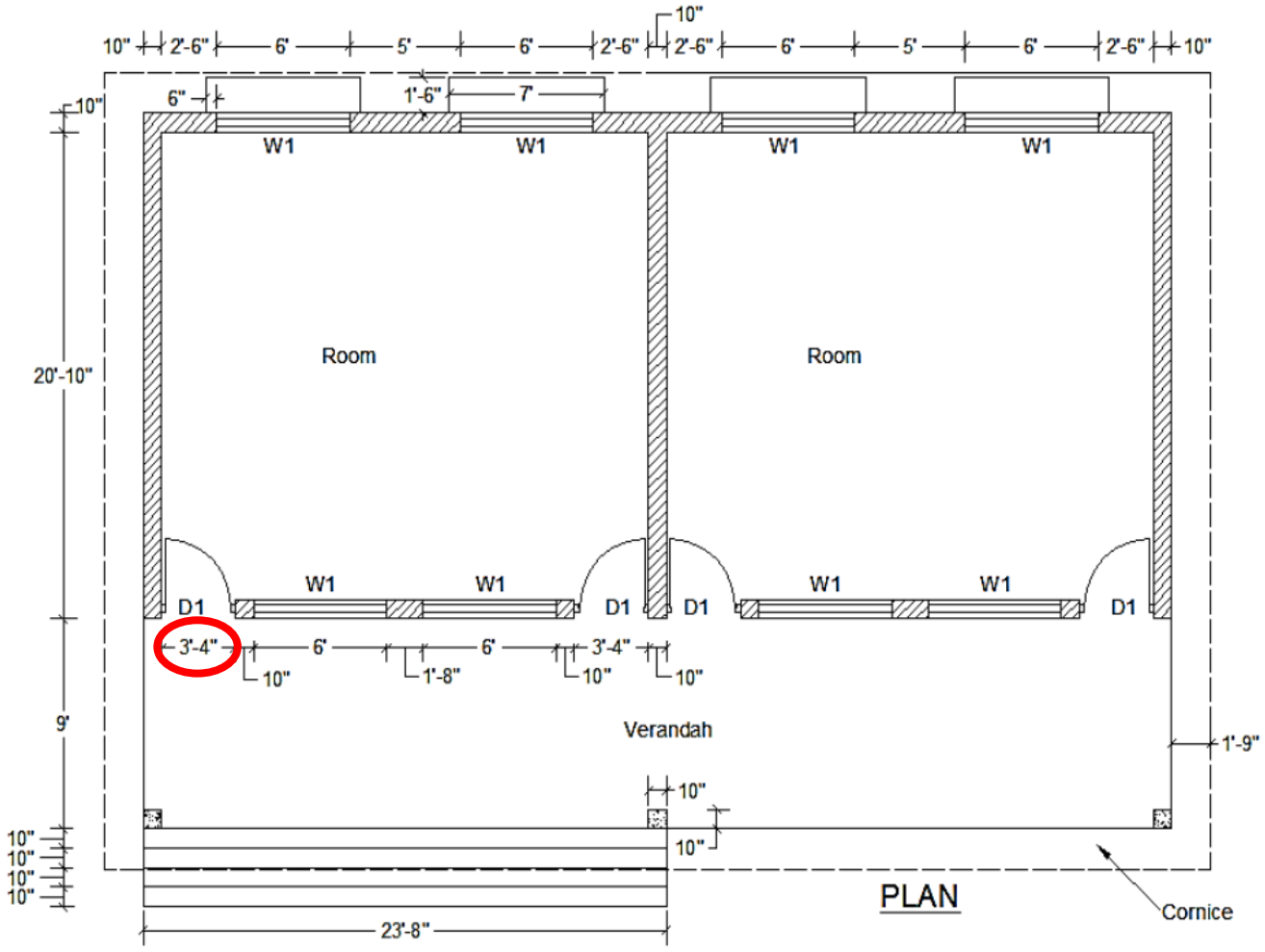
11. Floor Finish, ¼” thick



Verandah:

- Length= (22’10”)*2+10”
- Width= 9’
- Thickness= 1/4”

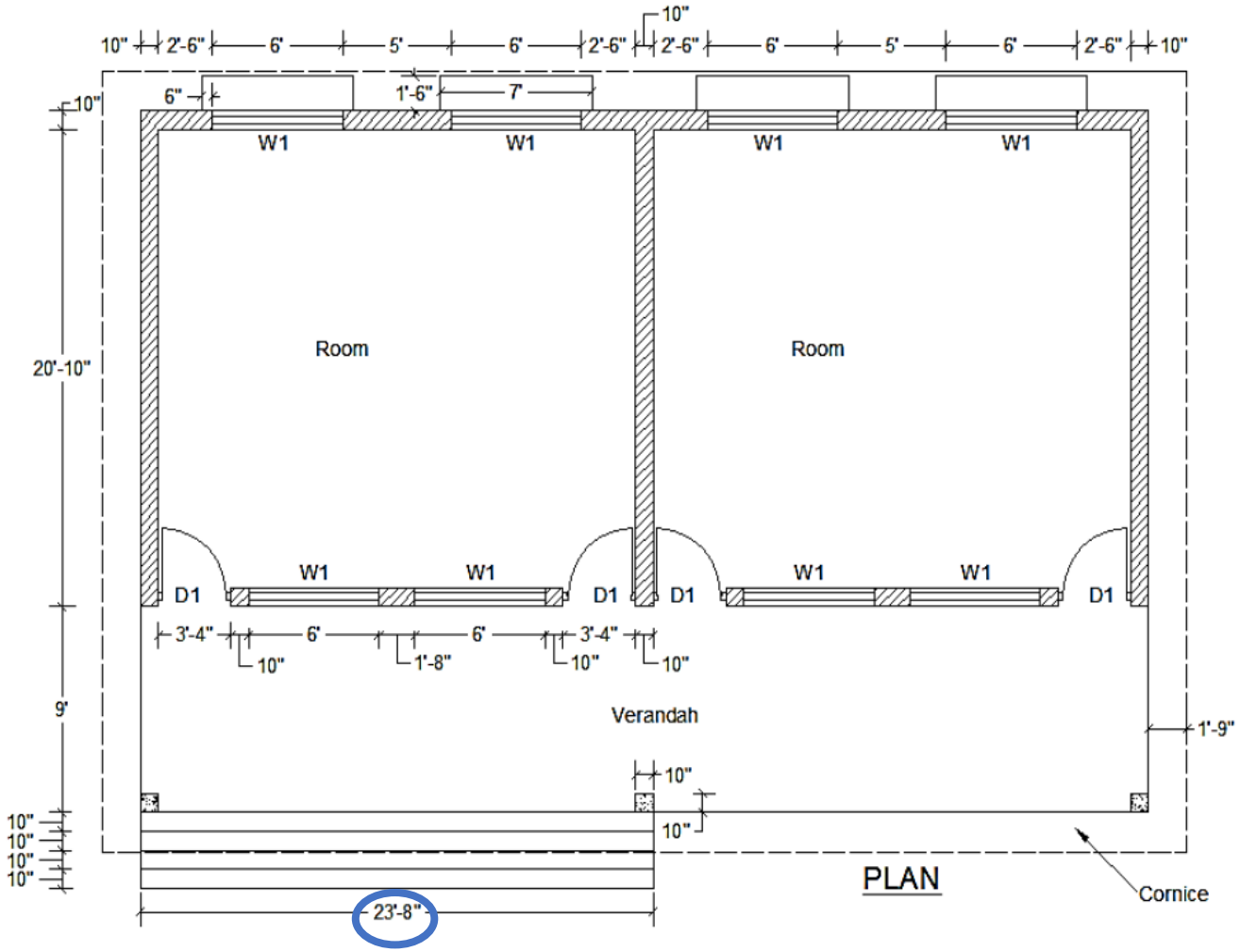
11. Floor Finish, ¼” thick



Door:

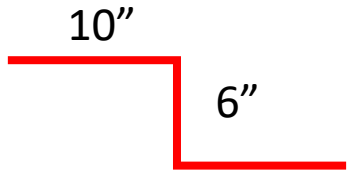
- Length= 3’4”
- Width= 10”
- Thickness= 1/4”

11. Floor Finish, ¼” thick

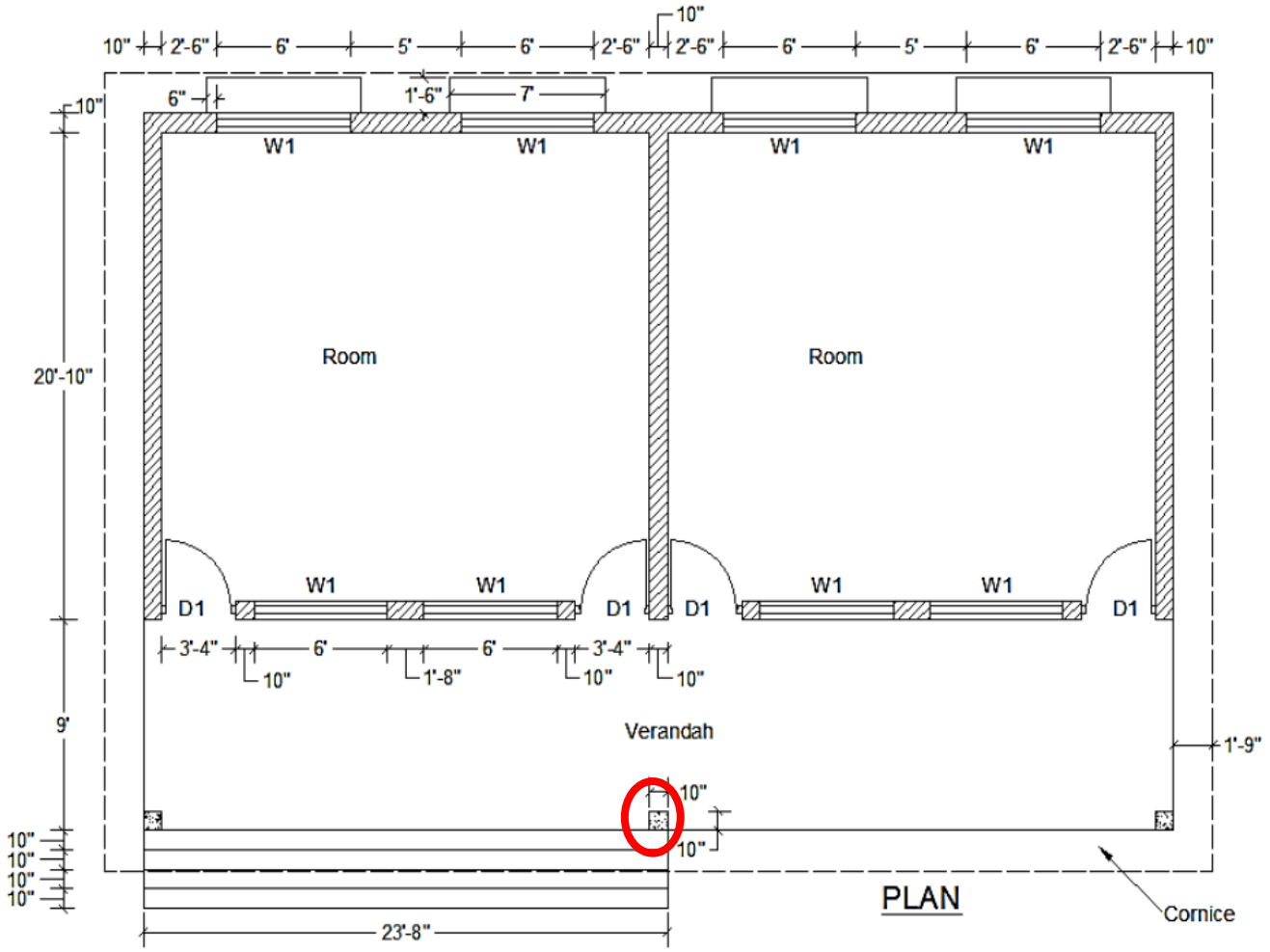


4 Steps:

- Length= 23’8”
- Width= 10”+6”
- Thickness= 1/4”



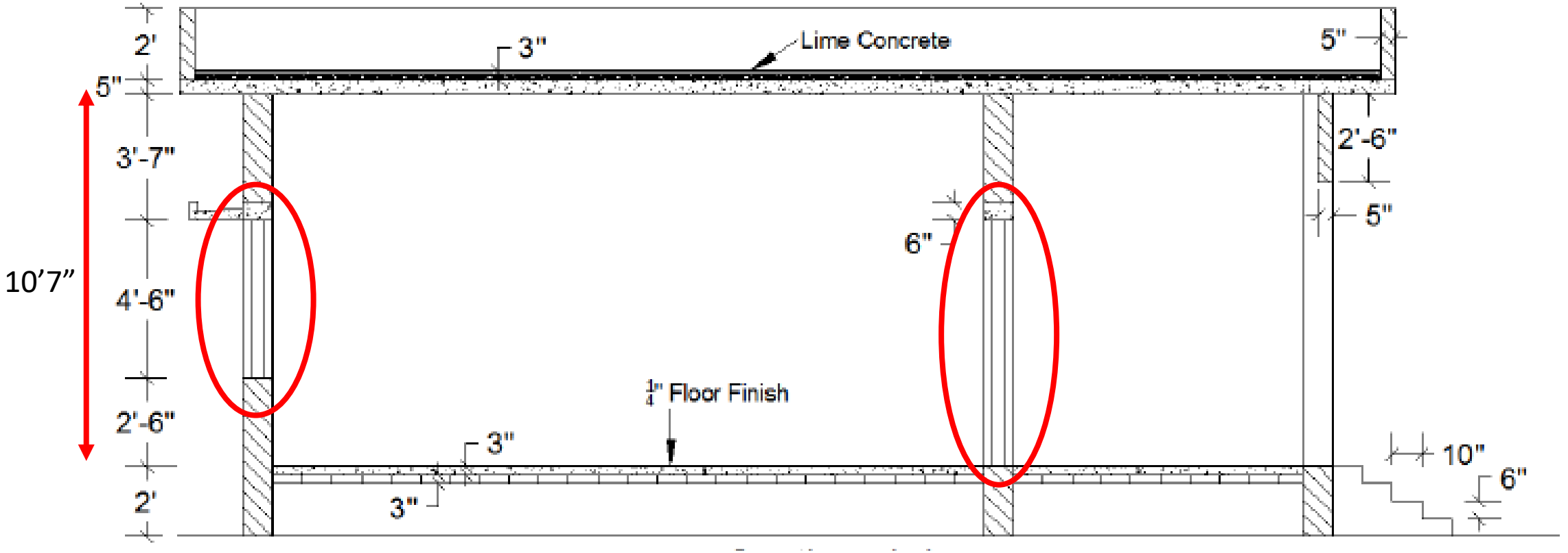
11. Floor Finish, ¼” thick



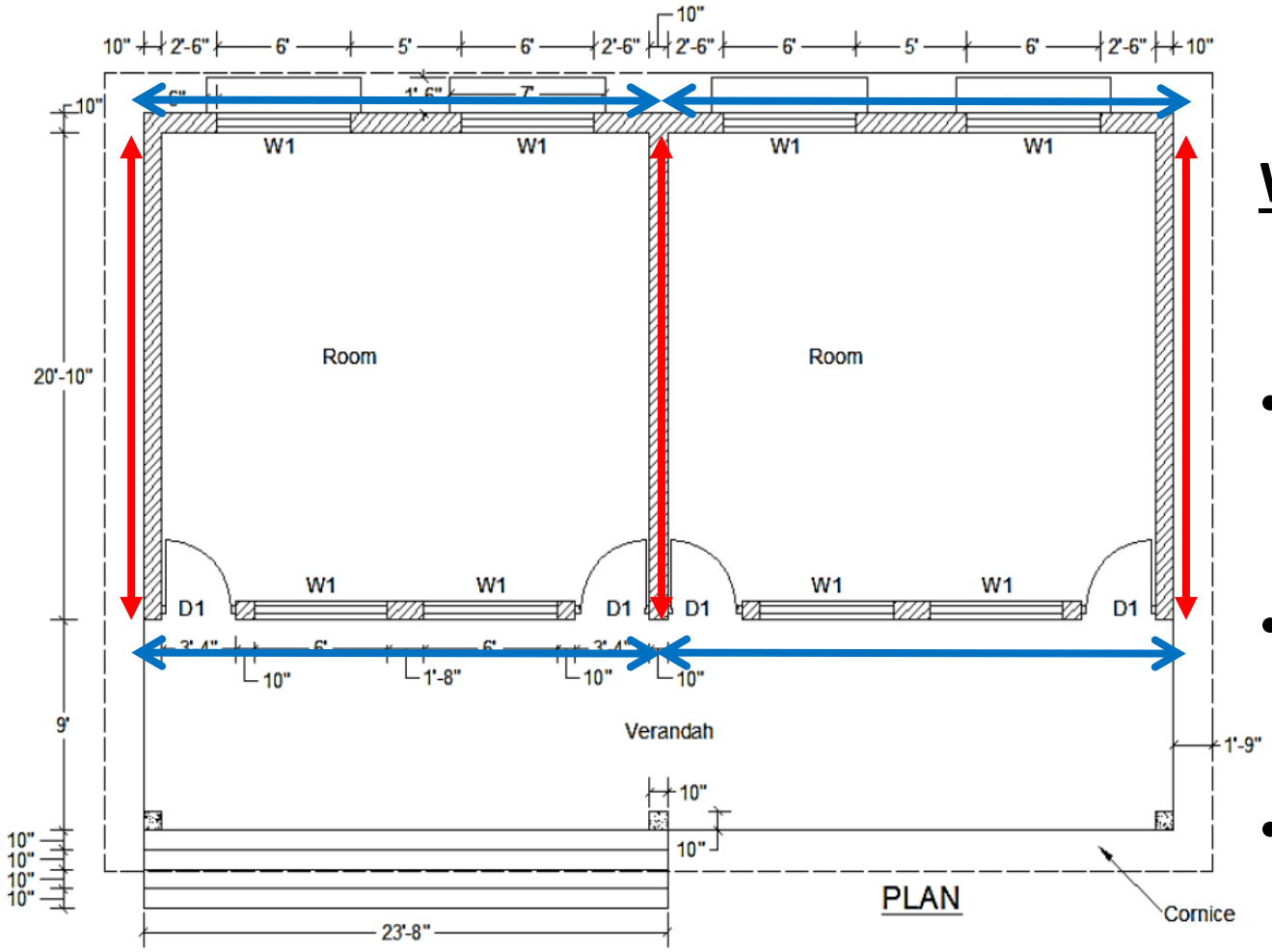
Deduct (Columns):

- Length= 10”
- Width= 10”

12. Brick walls in superstructure



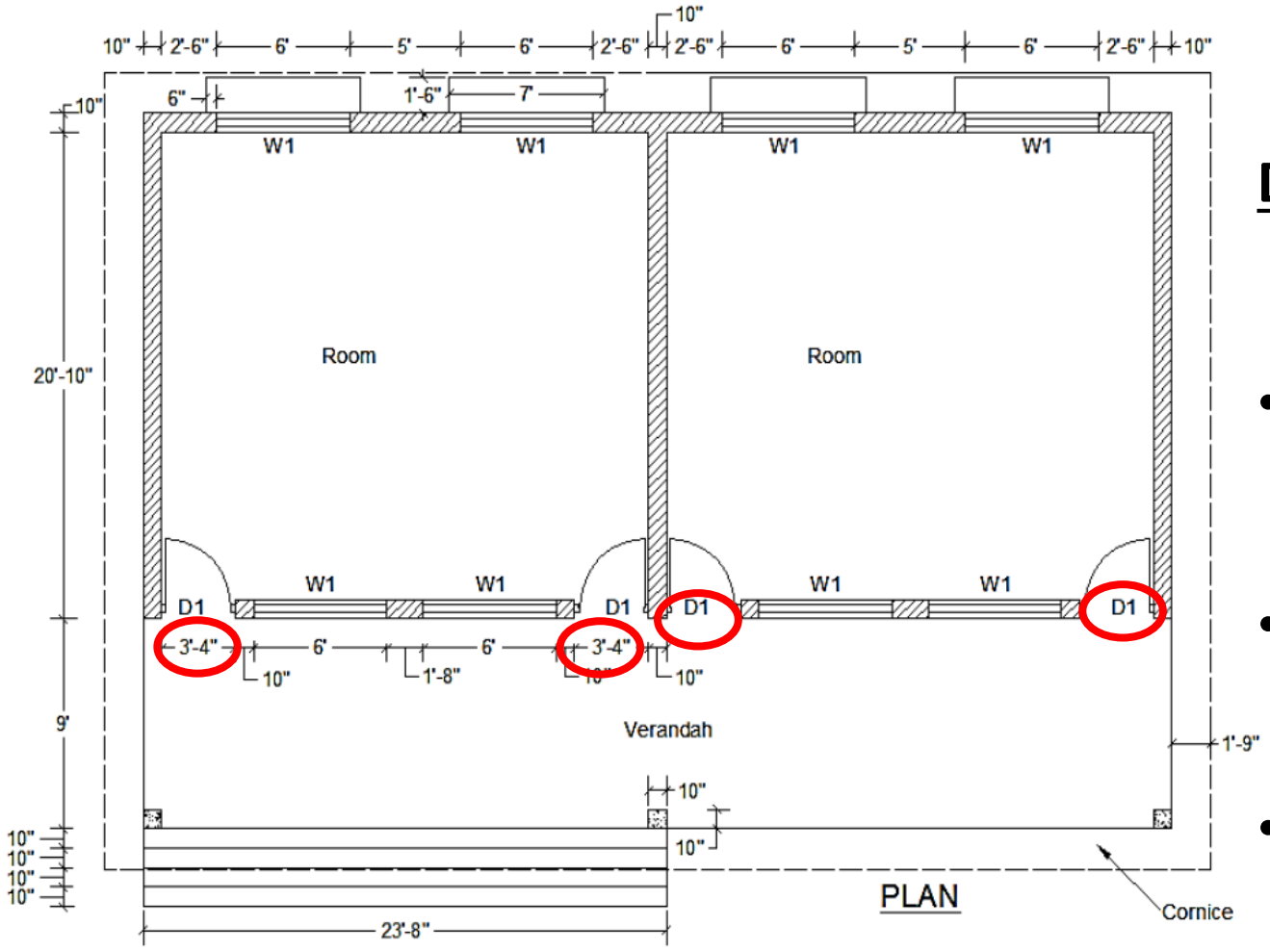
12. Brick walls in superstructure



Wall:

- Height= 10’7”
- Width= 10”
- Length= (20’10”-10”)*3 +(22’10”+5”)*4

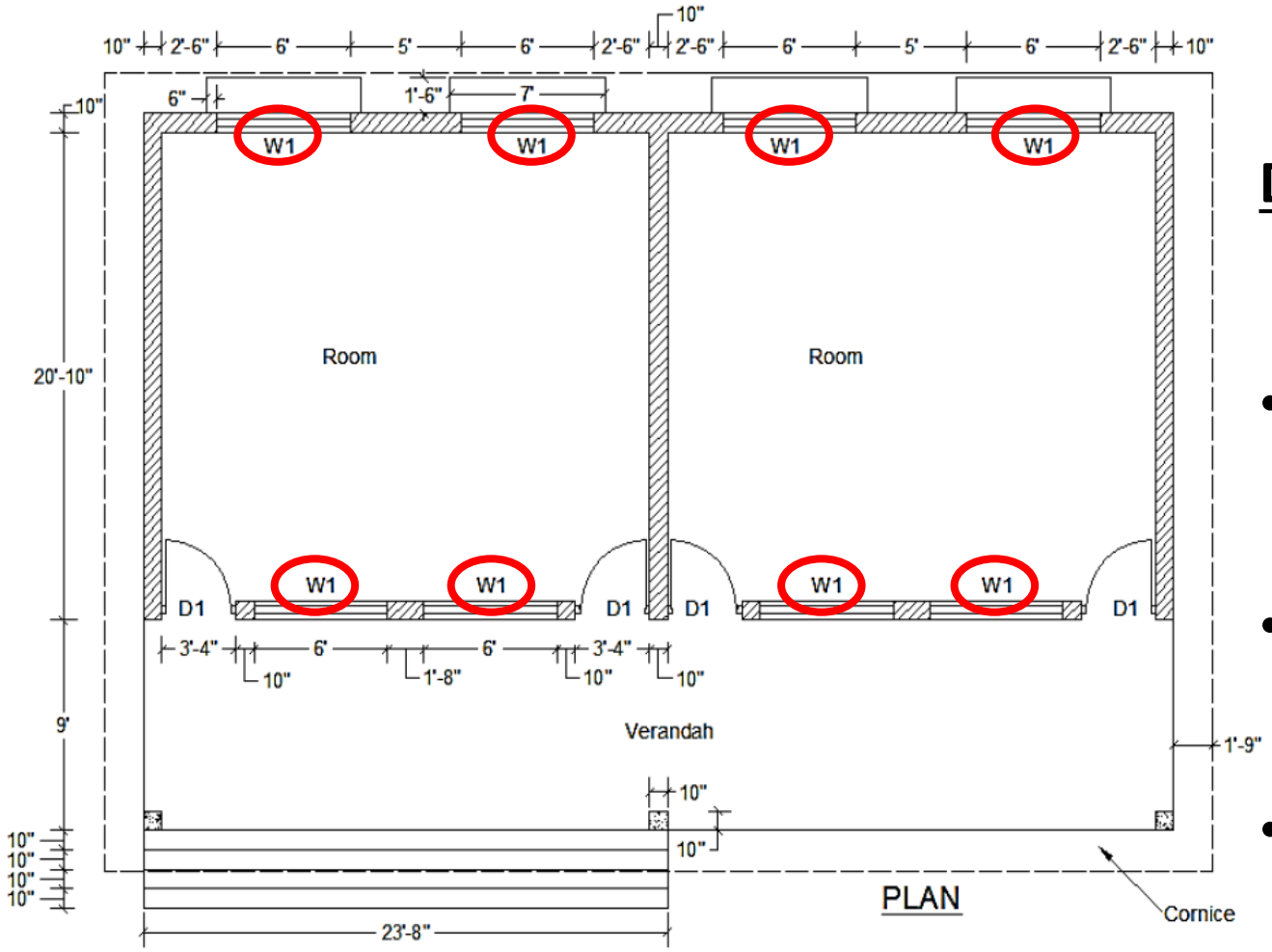
12. Brick walls in superstructure



Deduct (4 Doors):

- Height= 7’
- Width= 10”
- Length= 3’4”

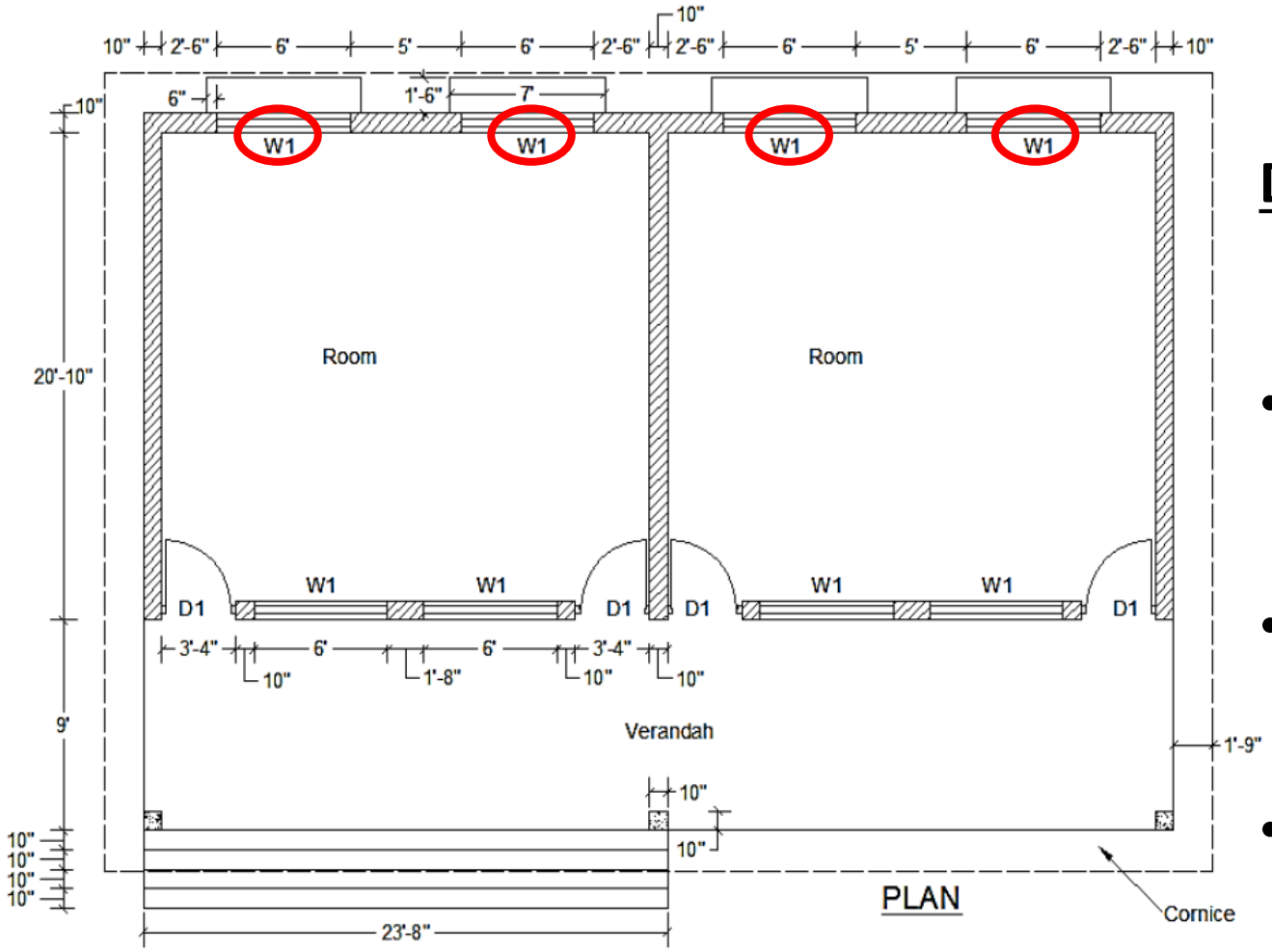
12. Brick walls in superstructure



Deduct (8 Windows):

- Height= 4’6”
- Width= 10”
- Length= 6’

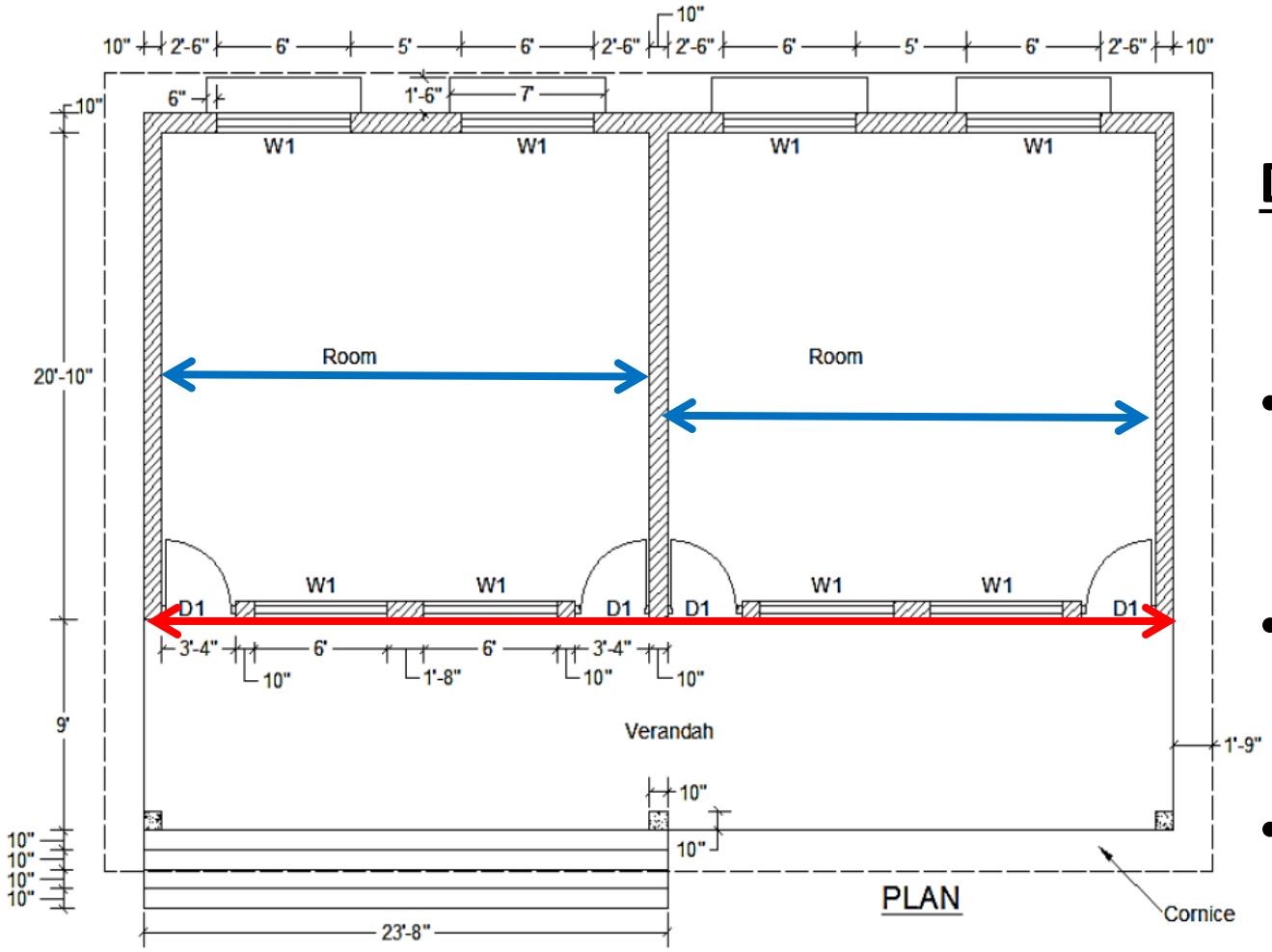
12. Brick walls in superstructure



Deduct (Lintel over Windows):

- Height= 6”
- Width= 10”
- Length= 7’

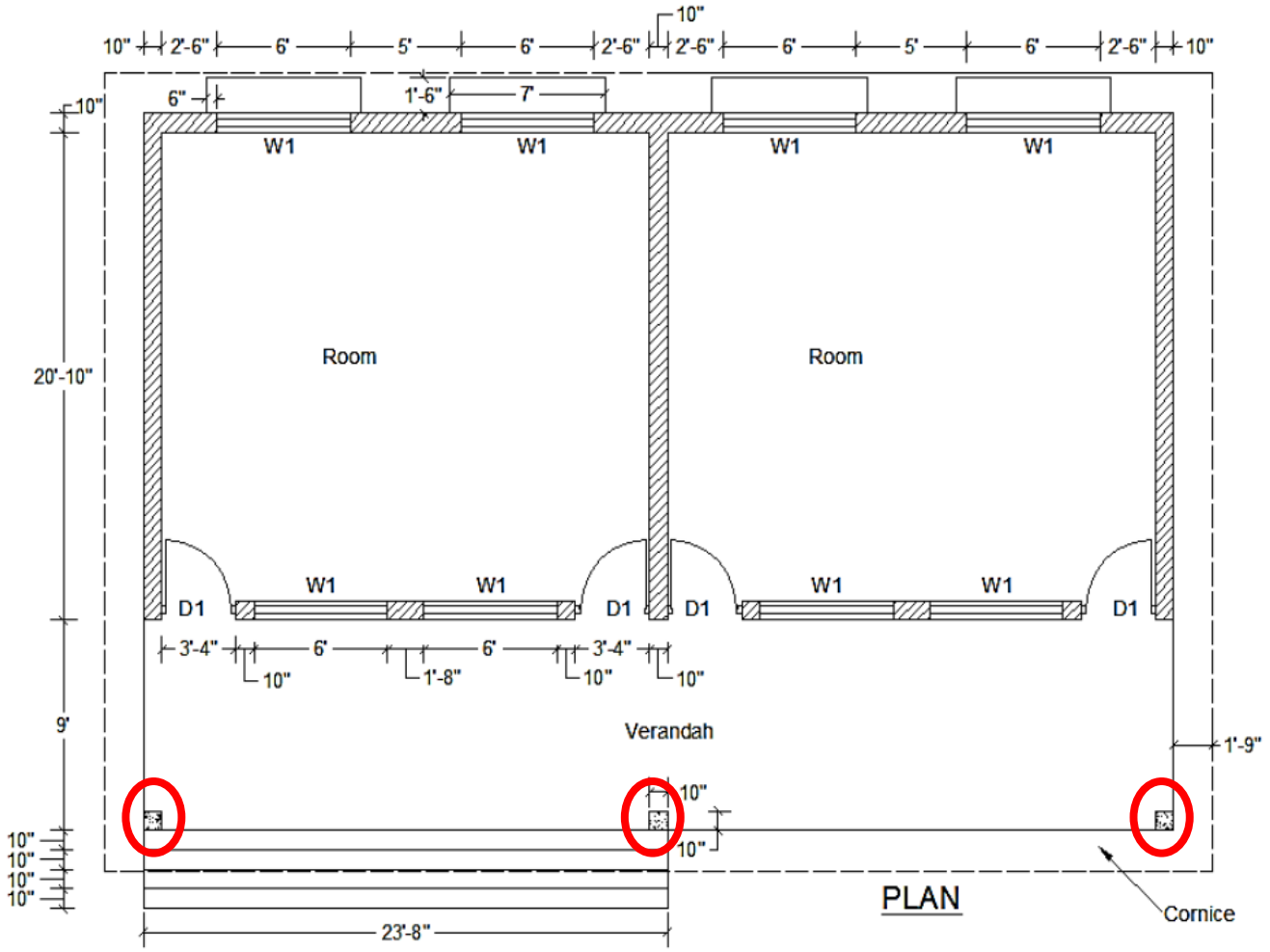
12. Brick walls in superstructure



Deduct (Lintel over W1+D1):

- Height= 6”
- Width= 10”
- Length= 2*22’+10”+2*6”

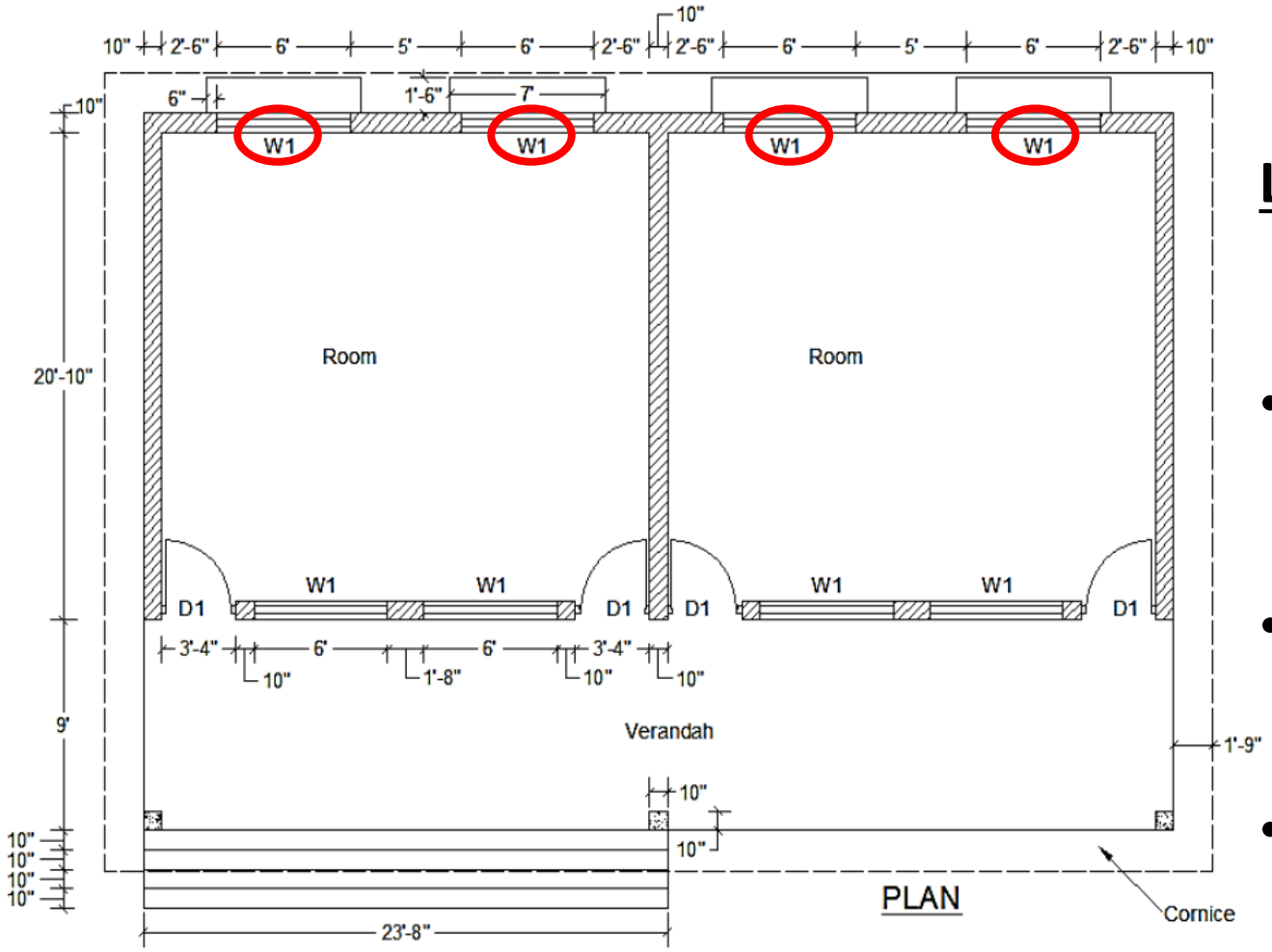
13. RCC in Columns



3 Columns:

- Length= 10”
- Width= 10”
- Height= 10’7”

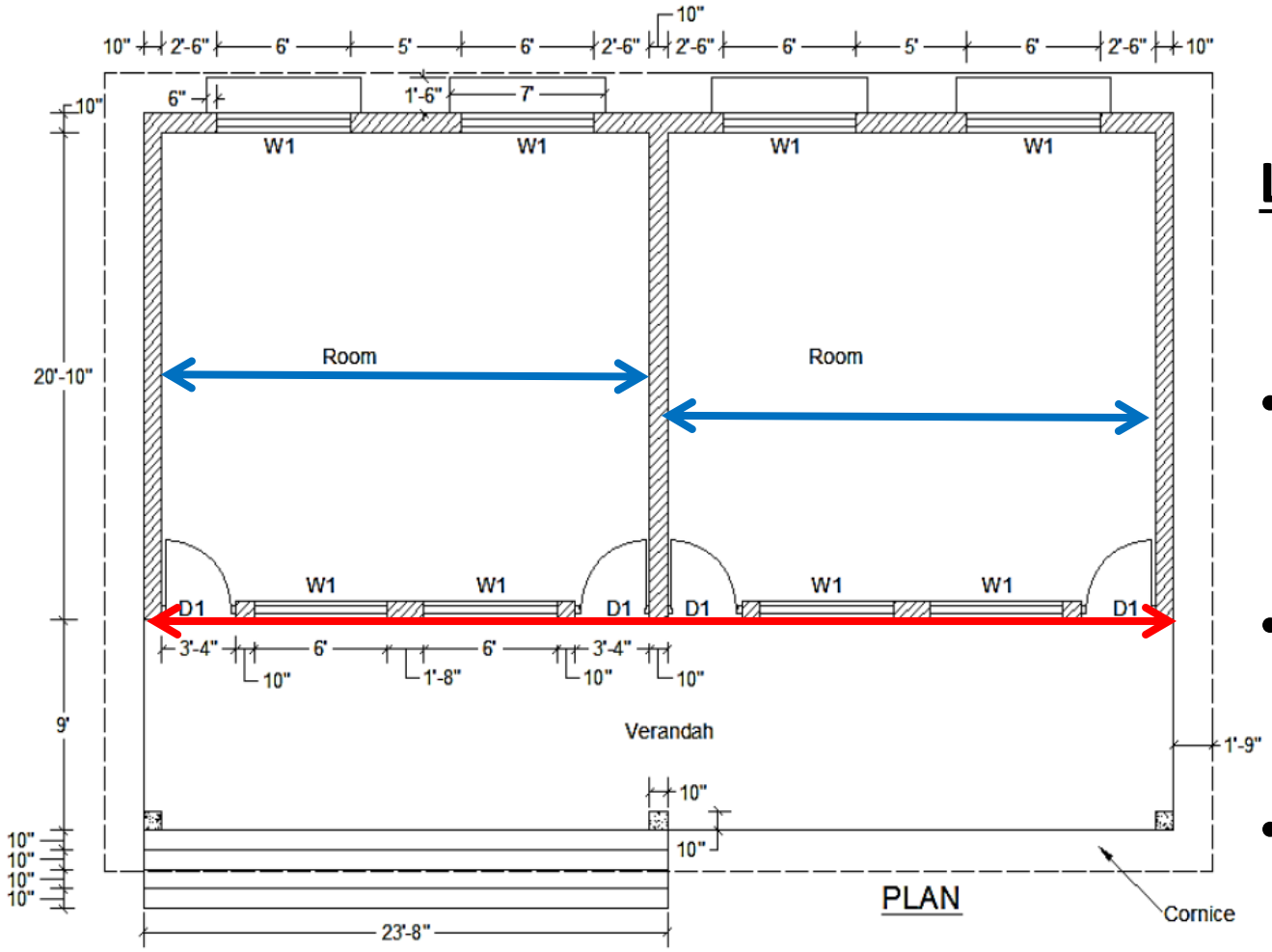
14. RCC in Lintel



Lintel over Windows:

- Height= 6”
- Width= 10”
- Length= 7’

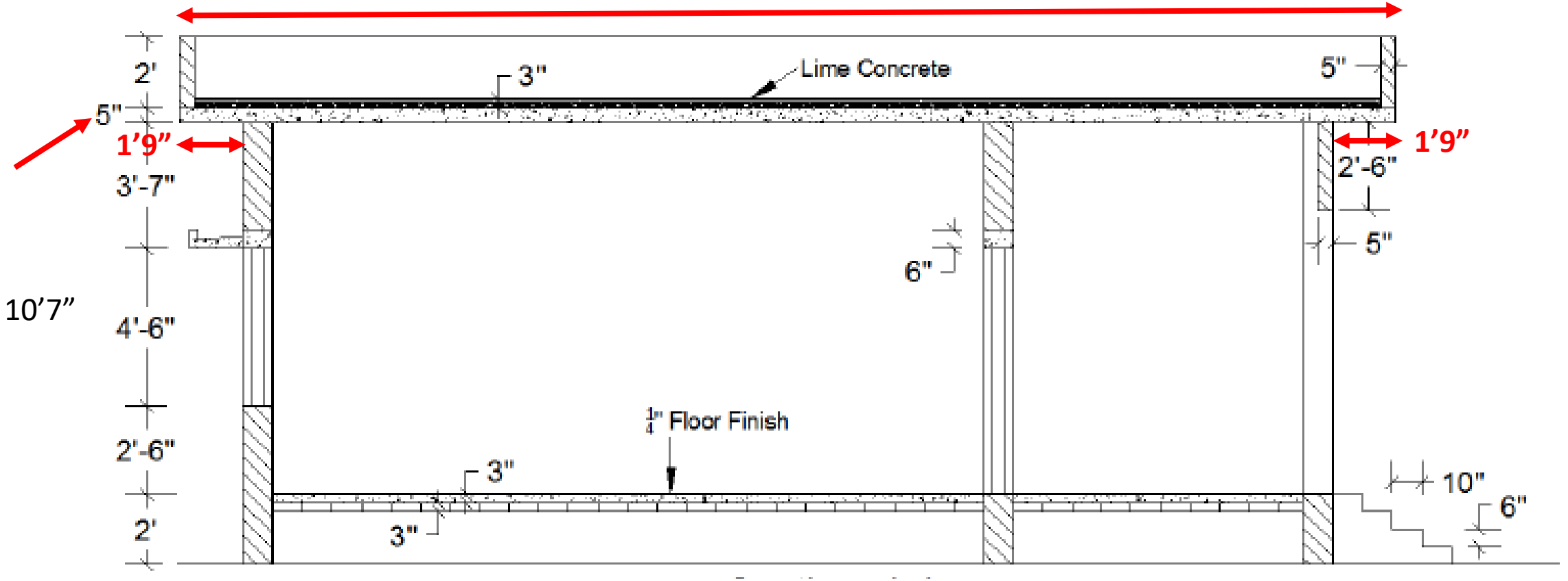
14. RCC in Lintel



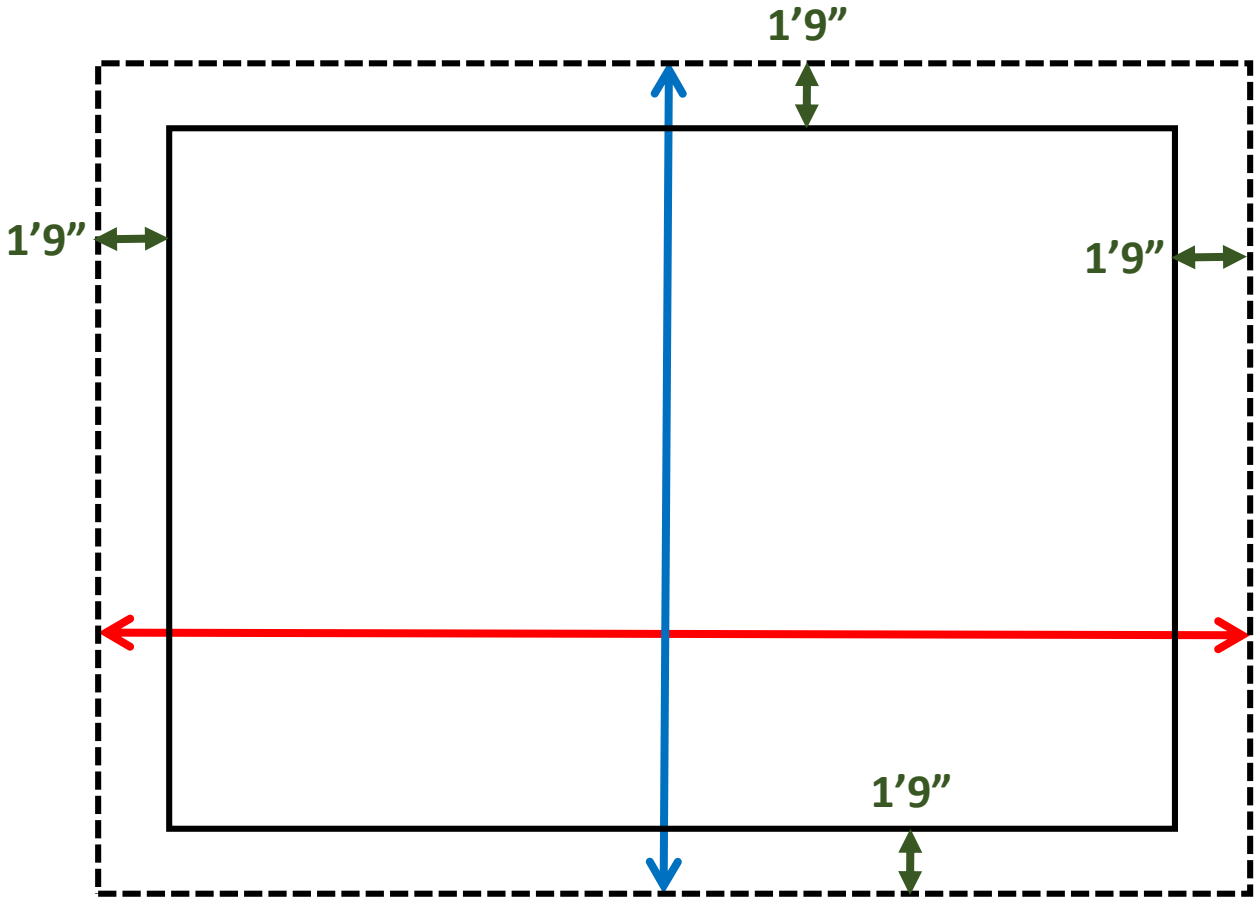
Lintel over W1+D1:

- Height= 6”
- Width= 10”
- Length= 2*22’+10”+2*6”

15. RCC in Roof

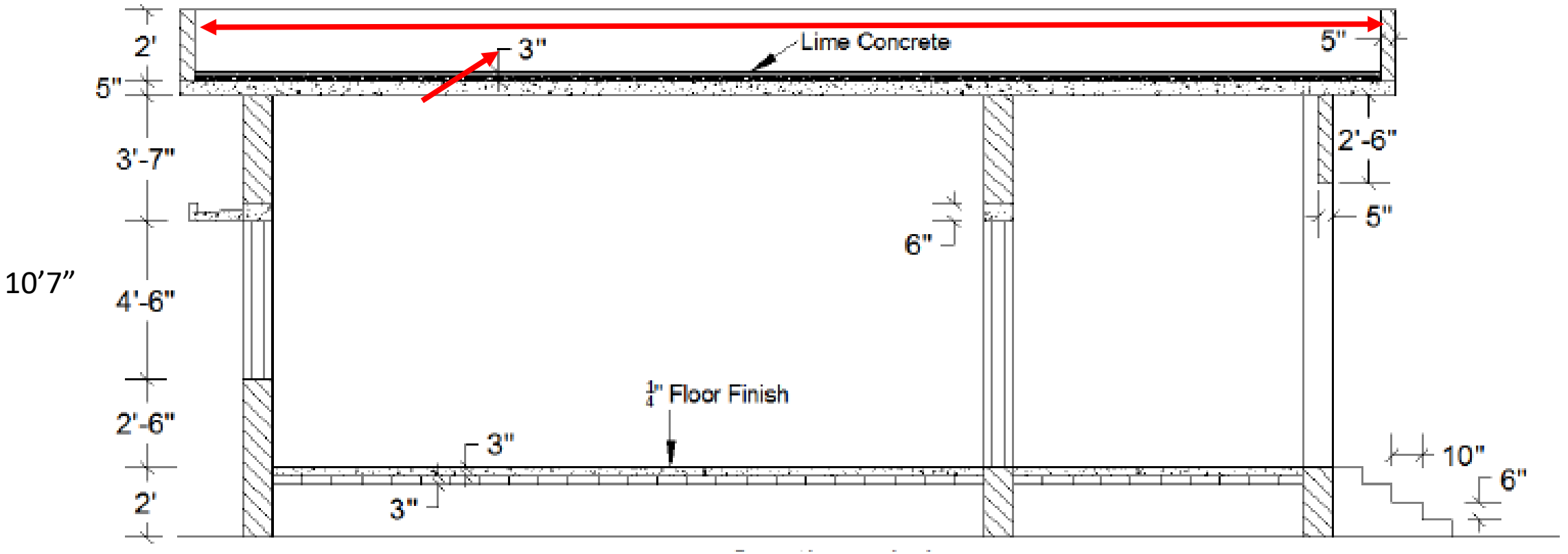


15. RCC in Roof

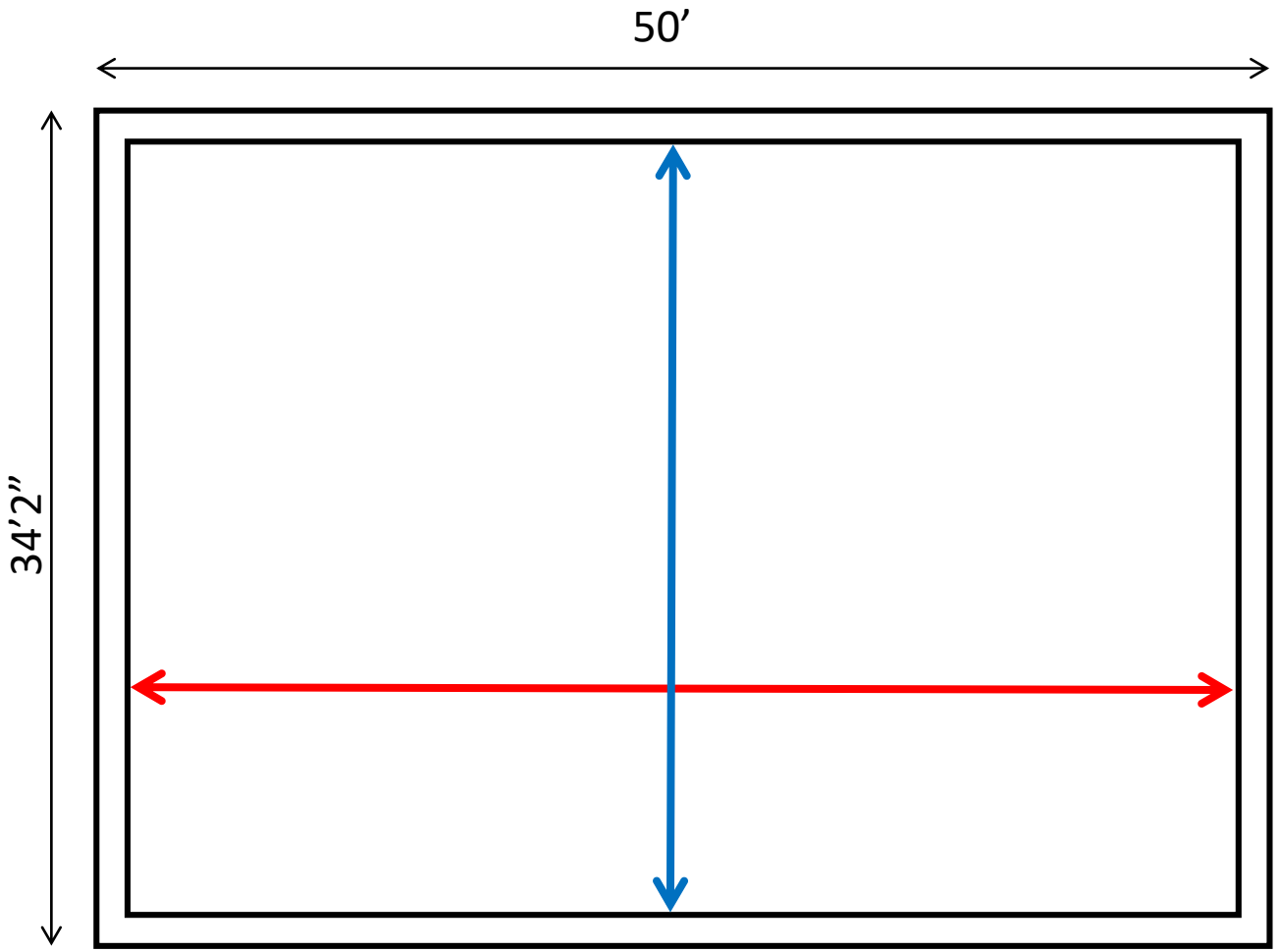


- Thickness= 5”
- Width= 30’8”+2*1’9”
- Length= 46’6”+2*1’9”

16. Lime Concrete in Roof

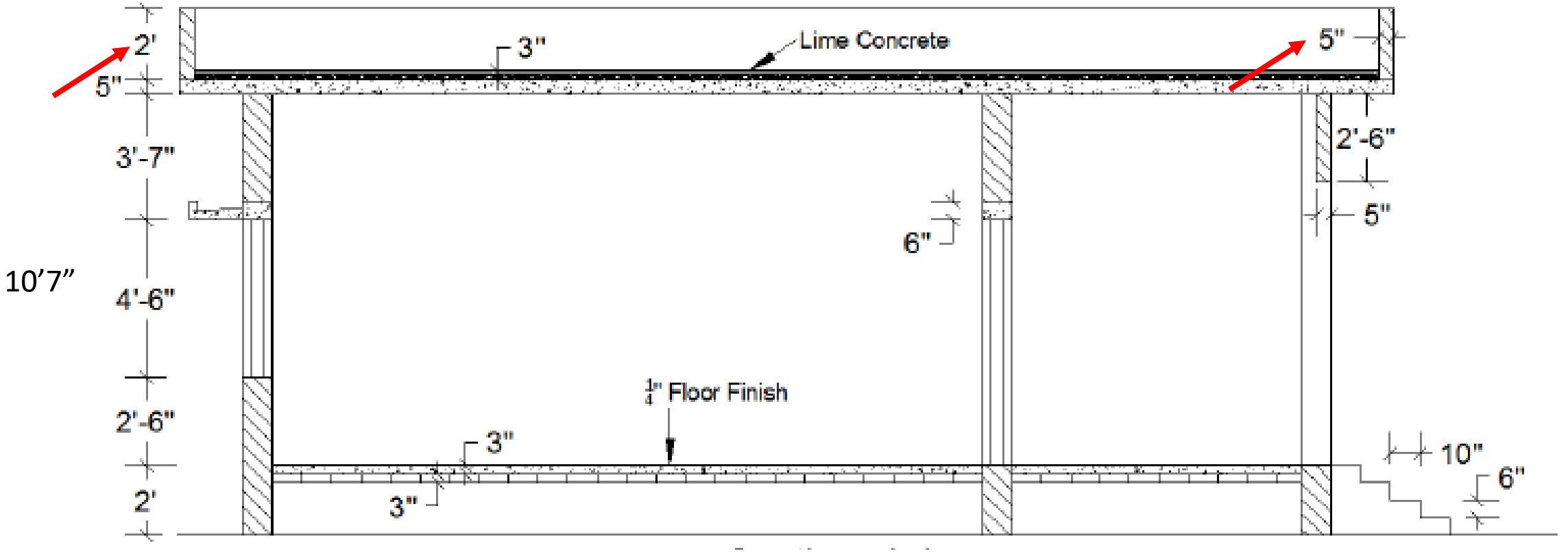


16. Lime Concrete in Roof

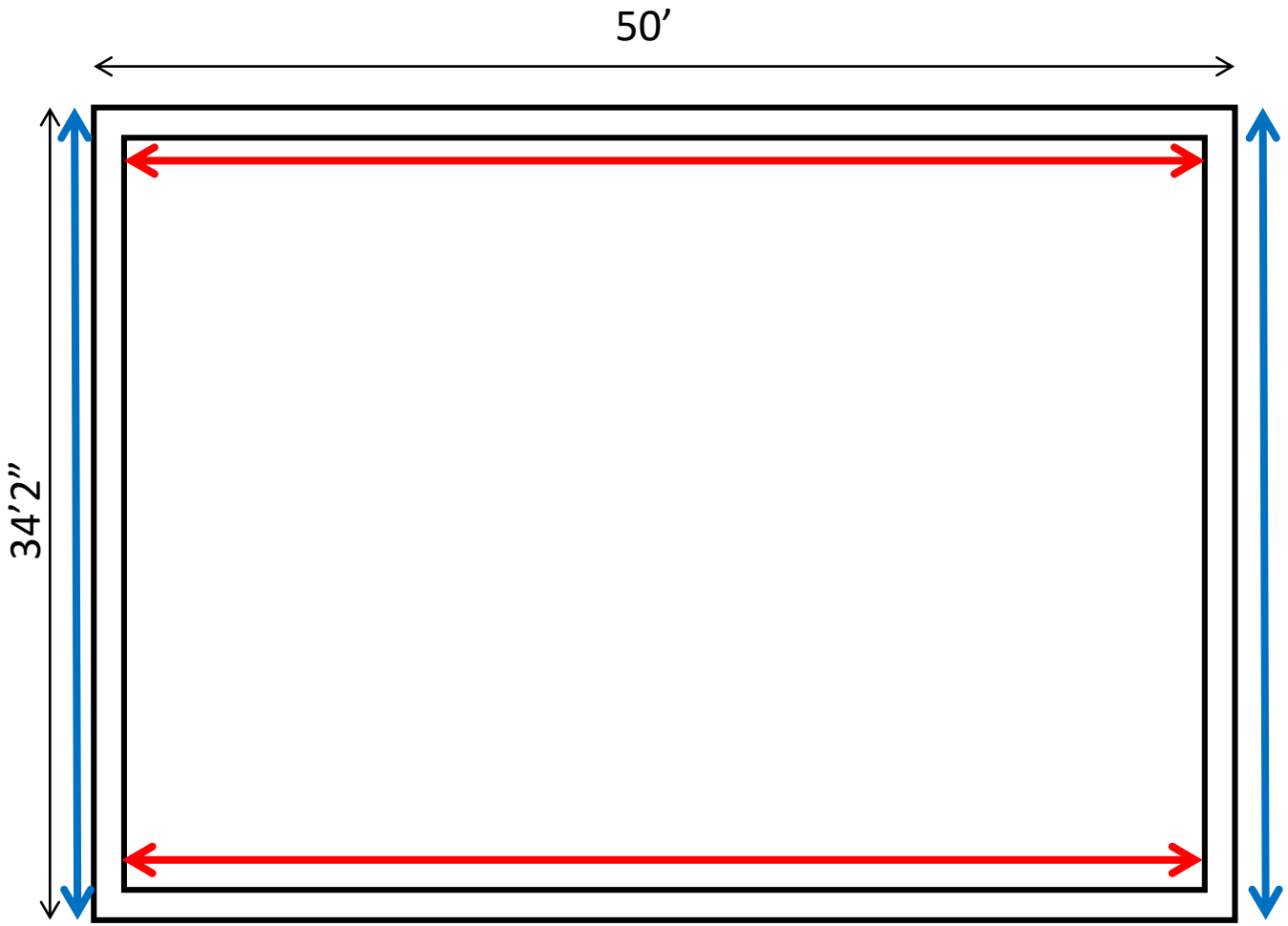


- Thickness= 3”
- Width= 34’2”-10”
- Length= 50’-10”

17. Bricks in Parapet

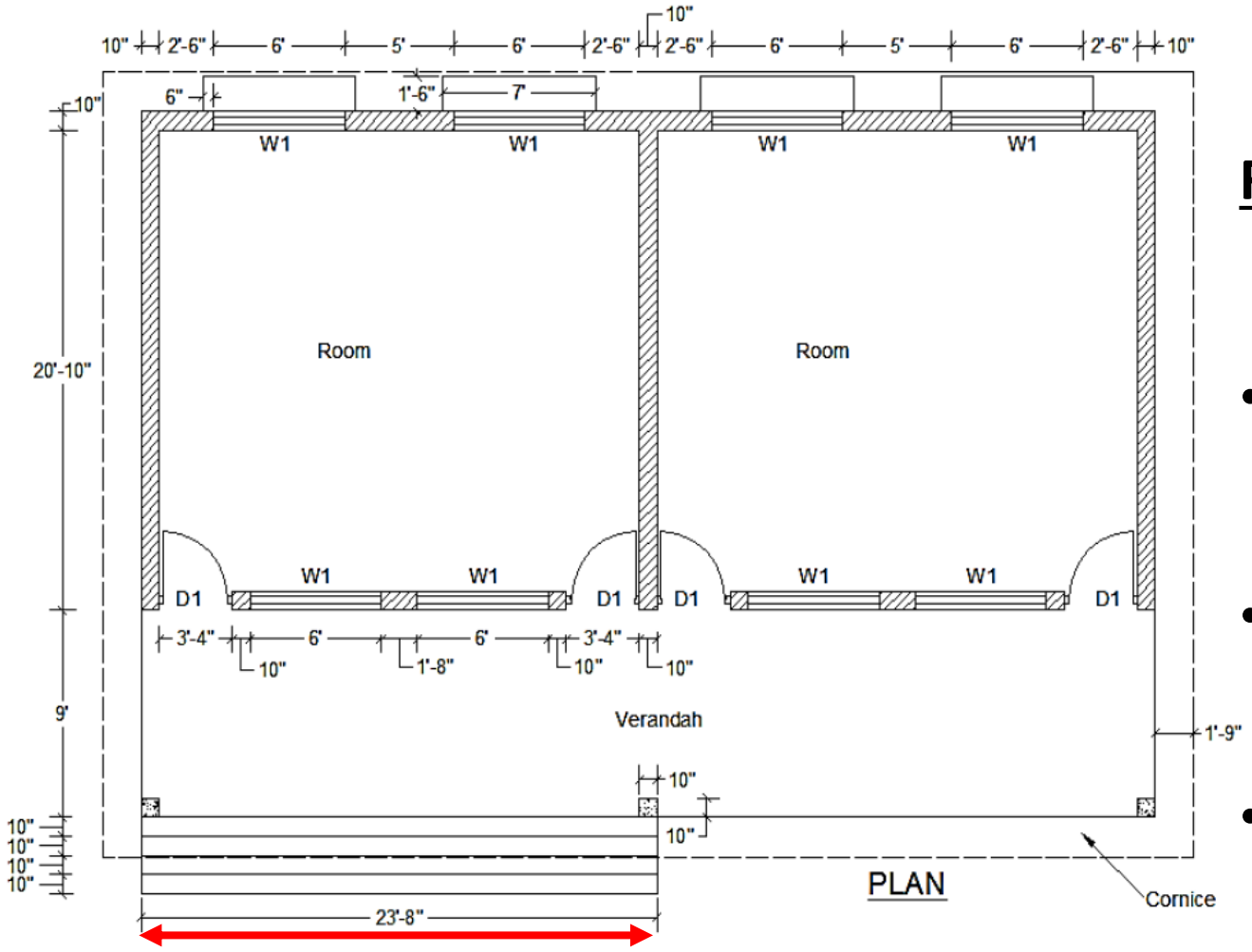


17. Bricks in Parapet



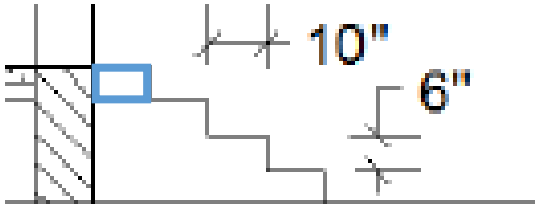
- Height= 2’
- Width= 5”
- Length= 2*((50’-10”)+34’2”)

18. Brickwork in Stairs

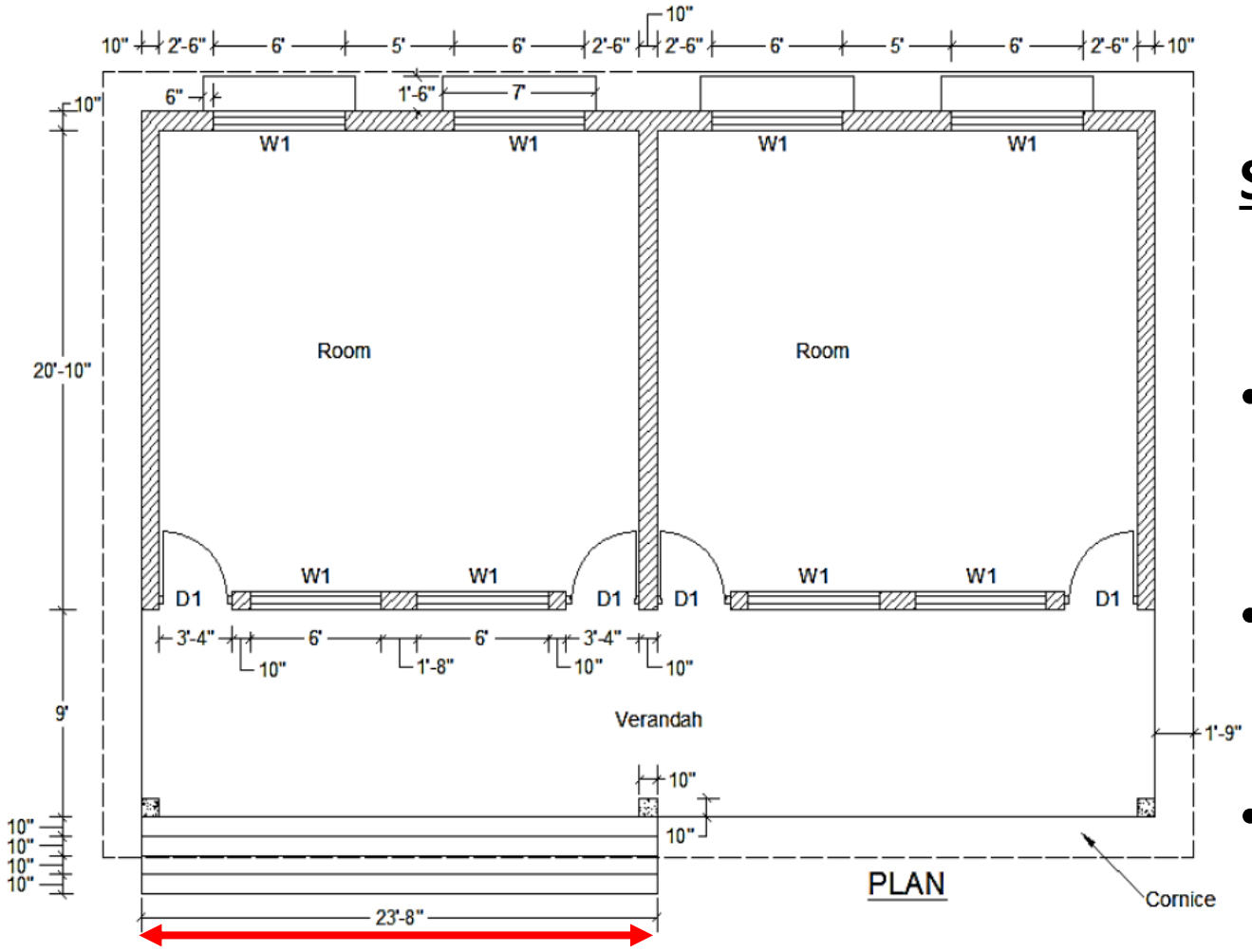


First Step:

- Height= 6”
- Width= 10”
- Length= 23’8”

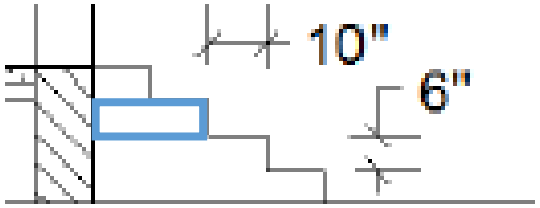


18. Brickwork in Stairs

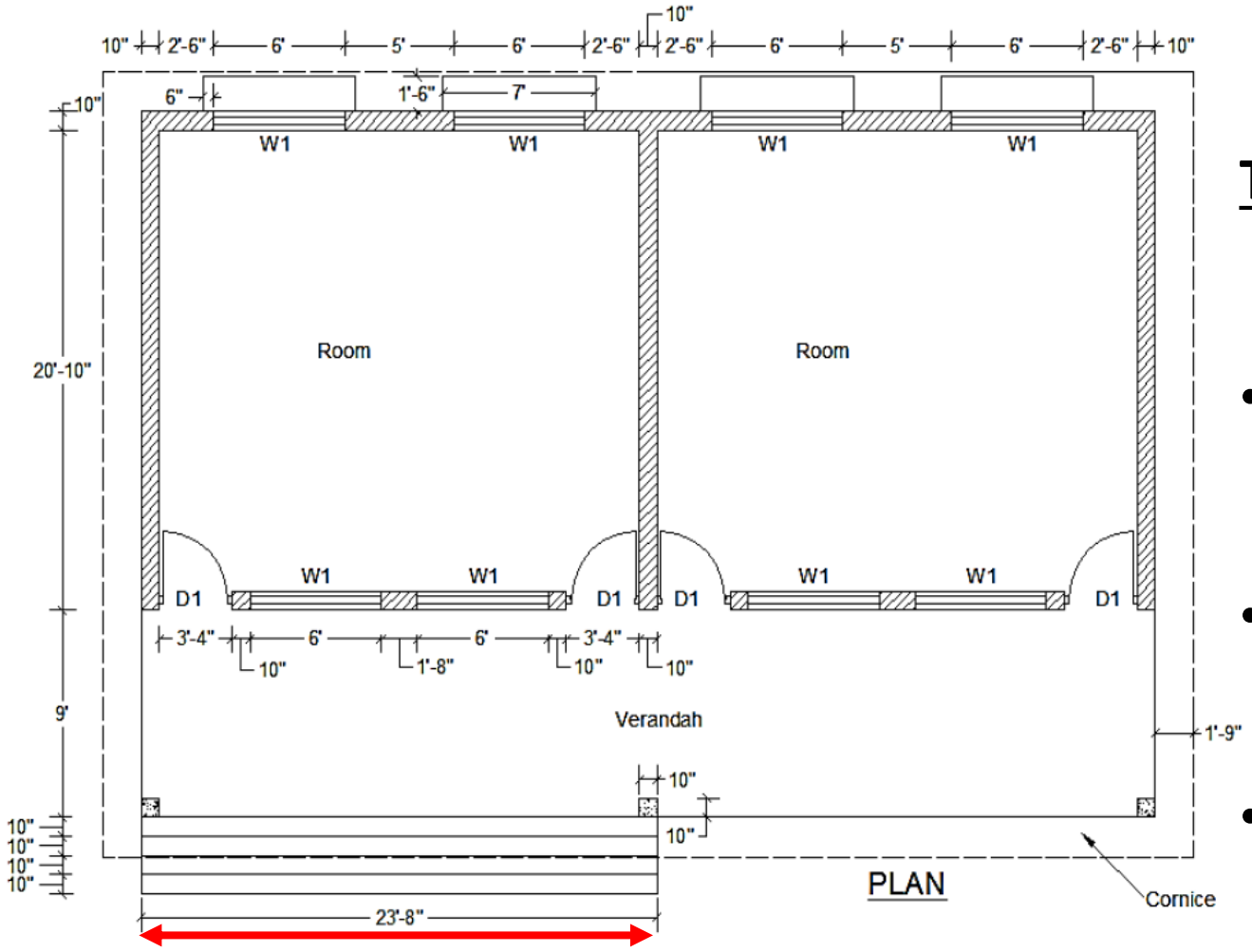


Second Step:

- Height= 6”
- Width= 20”
- Length= 23’8”

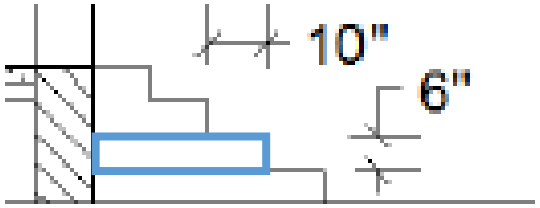


18. Brickwork in Stairs

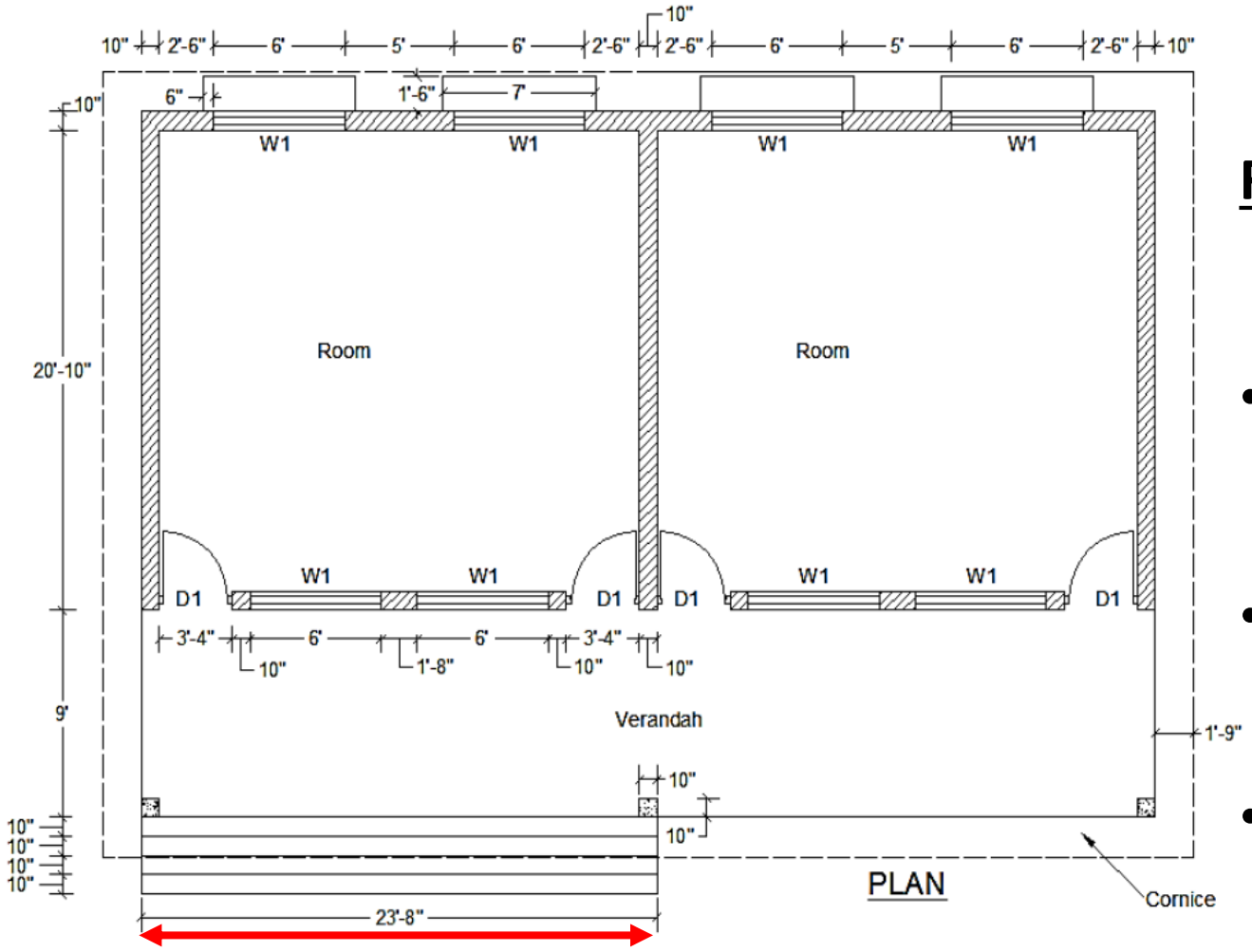


Third Step:

- Height= 6”
- Width= 30”
- Length= 23’8”

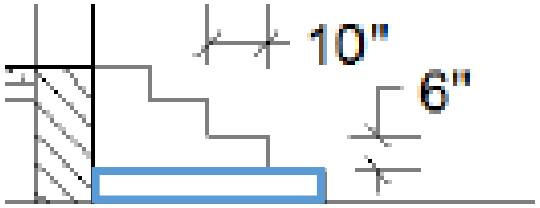


18. Brickwork in Stairs



Fourth Step:

- Height= 6”
- Width= 40”
- Length= 23’8”



Thank You!